NED University of Engineering and Technology
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GPU Sharing for Multi-Tenant Inference

A Bounded Systematic Evidence Synthesis of Performance, Isolation and Engineering Trade-offs

Operating Systems CEP – Topic 23
Task 3: Integrated Working Draft

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Working corpus: 28 report versions, 27 unique studies.
Frozen submission: 21 reports, 20 studies, preserved separately.
The bounded selection and remaining verification work are explicitly reported.

Abstract

GPU sharing for multi-tenant inference requires a choice among resource efficiency, request latency, isolation and implementation cost. This working review synthesizes 27 studies represented by 28 reports. The imported search register contains 4,189 occurrences and 4,075 canonical records after 114 duplicate links. In the continued 559-record branch, title/abstract screening excluded 225 records and advanced 334; seven additional studies met full-text eligibility, two reports were excluded at full text, and 325 advanced records await eligibility or identity closure. These branch counts do not constitute a reconciled final whole-review PRISMA flow. The analysis uses 270 appraisal items, 70 reported observations and two transparent calculations. It combines temporal scheduling, software spatial sharing, hardware partitions, runtime/API mediation, model-memory management and hybrid controls. The evidence supports conditional selection: fine-grained control can protect request latency, while model modification, profiling, transfer bandwidth and shared-context faults constrain its use. No pooled effect or universal winner is supported. The conclusions remain limited by incomplete selection, historical search-provenance gaps, title-only screening and source-version uncertainty.

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1 Introduction

1.1 Background and engineering problem

GPU sharing for inference is an operating-system resource-management problem: several services compete for finite execution capacity, device memory and access to a runtime that may not expose the control required by a service-level objective. Increasing concurrent execution can improve aggregate work completed while worsening the response of an individual service. Conversely, reserving the entire accelerator for one model can simplify interference management but leaves less opportunity to use spare capacity. The engineering question is therefore conditional: which mechanism should be selected for a particular workload, latency objective, memory requirement and isolation boundary?

The evidence makes this tension concrete. The ParvaGPU manuscript reports a local configuration in which throughput increases from 354 to 446 requests/s as the number of MPS processes rises from one to three, while reported latency increases from 11 to 27 ms. These outcomes move in opposite desirable directions. The latency quantile is unspecified, so the example cannot establish a P99 service guarantee. XSched instead reports a foreground P99 ratio of 1.07 relative to standalone execution, compared with 1.53 for native Triton in the recorded configuration. These are different engineering objectives, not numbers on one ranking scale [1, 2].

1.2 Scope and contribution

The unit of technical interpretation is concurrent inference on one physical GPU. A multi-GPU or cluster paper contributes only where a local sharing result can be separated from fleet scheduling. Mixed inference/training papers contribute inference outcomes and the associated background-work cost. Training throughput is not relabeled as inference throughput. Non-GPU accelerator results in otherwise relevant papers are outside the primary comparison.

This working draft integrates the frozen 20-study/21-report corpus with seven source-eligible additions, giving 27 studies represented by 28 reports. Its contribution is a decision-oriented comparison that joins mechanisms, evaluation conditions, reported outcomes, implementation costs and limitations. Existing surveys already examine multi-tenant GPU inference, accelerator-sharing efficiency/fairness/security, and GPU-datacenter scheduling [3, 4, 5]. The present taxonomy is therefore an organizational tool, not a first-survey claim. Its narrower emphasis is local GPU outcome traceability, comparability, source-version boundaries and conditional engineering decisions, including relevant 2025 systems. The survey audit records accessed versions and unavailable contextual leads; it does not establish exhaustive novelty. It distinguishes performance interference, resource partitioning, memory protection and fault containment. It also retains adverse outcomes rather than selecting only favorable headline speedups.

The review is not an exhaustive census of the search results. All 559 records in the continuation branch now have a title/abstract-stage decision: 225 exclusions and 334 advancements. Seven additions and two full-text exclusions are resolved; 325 advanced records still await eligibility or report-identity closure. This draft supports candidate selection under observed conditions while the wider coverage question remains open.

1.3 Objectives

The objectives are to classify the available mechanisms by their actual control boundaries; determine which outcomes are comparable; explain the conditions behind benefits and regressions; and derive recommendations that identify both expected benefit and sacrifice. The review does not seek a universal winning system. It seeks an auditable reason for choosing a candidate under stated constraints.

2 Research questions

The following research questions reproduce the submitted Task 1 protocol (Table 3).

RQ1 What time-slicing, MPS, MIG and API remoting mechanisms are evaluated for concurrent inference on one GPU, and how do sharing granularity, layer and isolation boundary classify them?

RQ2 How do studies measure utilization, throughput, density, average/tail latency, predictability and interference, and how comparable are their experimental methods?

RQ3 Under what GPU, workload, batch, tenant mix and configuration conditions are changes in utilization, throughput, density and latency reported?

RQ4 What resource and fault isolation properties, trade-offs, implementation costs and conflicting results are reported, and what limits the evidence?

RQ5 Which approach or combination is supported for specified inference-serving conditions and priorities, with what confidence and alternatives?

3 Research methodology

3.1 Protocol and implemented method

The submitted Task 1 protocol is preserved unchanged in the repository. It specifies independent screening and appraisal, stable record/report/study identifiers, page-level extraction and a condition-aware narrative synthesis. This draft follows its conceptual comparison rules while explicitly reporting deviations in execution. PRISMA supplies the reporting framework, while the search audit preserves the executed queries rather than replacing them with planned expressions [6, 7, 8].

The supplied Task 2 workbook is the source of the current corpus and historical decisions. GitHub's preceding status described an earlier 15-form assessment. Importing the newer workbook does not mean those older registers were silently rewritten. Task 3 stores the uploaded baseline separately and keeps the inherited research history available.

The implemented review uses a single automated source-based assessment under team direction. For the continuation, the user reported instructor permission to rely fully on this process; amendment M2026-10-23 records that oral permission as user-reported, rather than independently inspected written approval. No independent human item-level votes or

calibration are inferred. Activity records preserve actual preparation and decision times, which do not establish when a person reviewed a record.

3.2 Integrated source review of all included studies

The frozen source-review packages cover the original 20 studies and the REEF companion report: 70 appraisal items and 30 outcomes for Eshaal's allocation, 70 appraisal items and 13 outcomes for Raazia's, and 60 appraisal items and nine outcomes for Rameen's. All 200 appraisal items, 126 included-report inclusion-criterion checks and 52 retained outcomes have source-anchored review entries. The original baseline forms are preserved separately. The 12 inherited criterion checks for the two excluded reports are not recertified by this included-study review.

Corrections include GPU-specific VastPipe context; final iGniter anchors; gpulets' model-pair statistic; REEF conference/journal boundaries; Clockwork's TVM v0.7 environment; Torpor's evaluation/deployment distinction; KRYPTON's two-stage arrival-rate setup; XSched's Section 8 anchor; InferFair's goodput and slowdown-fairness definitions; ArkGPU's adverse result; Dyna-P's local sensitivity experiment; KRISP's AMD emulation; and Tally's total normalized system throughput. The supplied InferFair copy closes the last missing included-study source.

A common claim-specific Q9 appraisal distinguishes directly tested performance protection from untested security/fault guarantees. The gpulets artifact appendix also supports a higher reporting/reproducibility score without claiming local reproduction. The original 20-study scores sum to 338/396 applicable points; this aggregate describes reporting appraisal and must not be interpreted as an effect size or certainty rating. Original, source-reviewed and independent human fields remain separate. ParvaGPU numerical evidence is still specific to arXiv v1; final-text equivalence is not established. This integration does not supply independent human votes.

The continuation adds 42 source-anchored inclusion-criterion judgments, 70 Q1–Q10 items and 18 outcome rows for LithOS, Abacus, Miriam, SGDRC, Pantheon, Oneiros and GACER. The first six appraisal totals are17,17,16,16,16and17out of20; GACER scores13/18with a reasonedN/A isolation item. Combined with the original forms, the working corpus has 270 appraisal items, 168 included-report inclusion checks and 70 outcome rows. The study-level appraisal sum is 450/534 applicable points. This accounting describes reporting quality, not independent replication or a certainty-weighted effect. The seven additions and their anchors appear in Appendix H [9, 10, 11, 12, 13, 14].

3.3 Databases, search fields and period

The active occurrence register represents ACM Digital Library, IEEE Xplore, Springer Nature Link and ScienceDirect, with supplementary named-paper lookups recorded separately. Their relevance is complementary: systems conferences, parallel/distributed systems research and journal literature provide different routes to sharing mechanisms. However, representation in an imported register is not equivalent to verified complete coverage of every planned database run.

The main publication window is 2020 through the date of each executed search in October

2026. IEEE and Springer entries record 11 October; ACM and ScienceDirect entries record 14 October. Later retrieval of a paper does not change an earlier search cutoff. The older journal survey used as an organizational benchmark is contextual literature, not an additional included primary study.

The executed expressions generally combine GPU terms, inference/model-serving terms and mechanism terms. Some interfaces use broad indexed fields rather than title/abstract/keywords. Broad fields can retrieve papers in which sharing terminology occurs only in references or background. Conversely, restricted inference expressions can miss papers described only as LLM serving. The exact retained expressions, fields and filters are reproduced in Appendix A; the workbook holds filenames and provenance.

The historical ACM master record contains displayed, claimed and parsed totals of 1,600, 1,603 and 1,829. The ScienceDirect MPS record retains 517 displayed against 563 registered. These differences are not corrected by inventing a missing export or changing a count to match another artifact. The active consolidated occurrence totals and their original discrepancies are different fields.

3.4 Eligibility and study identity

All inclusion criteria must be met: an English peer-reviewed primary report; evaluated concurrent sharing on one physical GPU; inference outcomes, separable where training is also present; interpretable outcomes and configuration; sufficient local technical evidence; and the specified publication period or justified older exception. A paper's title or prestigious venue alone does not establish eligibility.

Decisive exclusion categories are non-primary/non-peer-reviewed material, no evaluable inference-sharing mechanism, hardware-only work without evaluated systems sharing, aggregate-only multi-GPU evidence, non-GPU scope, insufficient assessed technical detail, and period/language mismatch. The supplied source assessment excludes TGS and HiveD because the extracted evaluation concerns training rather than eligible inference-sharing outcomes. These exclusions are kept outside the included-paper catalogue.

Records, reports and studies are distinct entities. A repeated database occurrence is linked during deduplication. A report version remains identifiable even when it belongs to the same underlying study as another report. REEF's conference report P005 and journal extension P023 map to S005; the journal's first page explicitly states the extension relationship [15, 16]. They cannot supply two independent replications. Other manuscript copies retain their version labels rather than automatically inheriting final-publication page numbers.

3.5 Appraisal and extraction

The Q1–Q10 rubric scores 0, 1 or 2 for objective clarity, mechanism description, experimental transparency, hardware/workload description, baseline fairness, utilization/throughput evidence, latency measurement, multi-tenant interference, isolation evaluation and replication/limitations. Justified not-applicable items reduce the applicable denominator. Scores are reporting and internal-validity appraisals, not journal-prestige scores or automatic exclusion criteria.

There are 270 appraisal items for 27 study-level forms: 200 inherited items and 70 new

source-based items. After the source-based denominator review, two items are N/A: P001 Q7 and P006 Q9. XSched Q9 is applicable to its directly evaluated performance protection. The frozen baseline retains its original three N/A items. The normalized score is the recorded sum divided by the applicable maximum. The workbook recalculates these sums from the item rows and checks them against the study table. A companion report does not create an additional independent study-level appraisal.

Extraction preserves mechanism, environment, workload, tenants, load/SLO, baseline, result, isolation dimensions, overhead, limitations, source and anchor. The 70 outcome rows and two calculations are not 68 studies. Missing values remain missing or explicitly not established. A throughput ratio is not converted into utilization percentage, a model-pair distribution is not treated as a request-latency percentile, and a reported maximum is not recast as an average.

3.6 Synthesis and robustness rules

The synthesis is narrative and comparative. A numerical contrast is interpreted within the source experiment unless GPU, model, batch, load, concurrency, metric and baseline are sufficiently aligned across studies. No pooled speedup or weighted cross-study average is calculated. A result is first evaluated for relevance to the decision, then for comparability and appraisal limitations.

The protocol's below-60% sensitivity view removes no current study because all recorded normalized scores are at least 70%. Consequently, an unchanged answer under this cutoff is mechanically expected. It does not demonstrate robustness to publication bias, title-only screening, a different stopping rule or correlated single-reviewer error. Claim-level anchors and version checks remain necessary even for high-scoring studies.

4 PRISMA study selection and corpus

The frozen selection logs reconcile 4,189 occurrences minus 114 duplicate links to 4,075 canonical records. Their historical partition is 3,493 screening exclusions, 559 retained records not advanced, 21 included reports and two source-assessed exclusions. That snapshot is preserved separately and is not overwritten by the continuation.

The 559-record branch now has final title/abstract-stage decisions: 225 excluded and 334 advanced. A conservative scope audit reopened 56 earlier proposed exclusions. Of the screening exclusions, 108 explicitly use title/metadata evidence alone. No record is excluded because an abstract is missing. The queue contains 235 original abstract texts plus 161 supplemental recovered texts; 163 records still lack abstract text, and five recovered texts have quality flags.

Of the 334 advanced records, seven new reports met all inclusion criteria and two were excluded at full text under EC2. In-Mig evaluates cross-venue agent reasoning and routing without an extractable GPU-sharing intervention; ALL-RP evaluates social-media repost prediction, with GPU use for model training rather than concurrent inference sharing. The remaining 326 records require eligibility or identity closure. They are neither full-text exclusions nor reports declared unavailable. Duplicate and companion-report proposals remain separate identity checks.

The working included set is therefore 28 reports representing 27 studies, preserving the REEF P005/P023-to-S005 linkage. These counts do not by themselves finalize whole-review PRISMA: historical record/report mapping, source-assessment events and pending identities must be reconciled before a complete final flow is claimed. Later retrieval dates do not change the original search cutoffs. The minimum-study target is met; it is not a cap or a reason to omit further eligible studies.

Accounting scope / stage	**Count**
Historical identified occurrences	4,189
Historical duplicate links	114
Historical canonical records	4,075
Historical original included reports / studies	21 / 20
Continuation records screened	559
Continuation screening exclusions	225
Continuation advanced to retrieval / eligibility	334
Continuation full-text inclusions / exclusions	7 / 2
Advanced records with unresolved eligibility or identity	326
Working included reports / unique studies	28 / 27

Stage distinctions: $559 = 225 + 334$ and $334 = 7 + 2 + 325$. Advancement does not mean that all 334 reports have been retrieved or assessed. This is branch accounting, not a completed whole-review PRISMA flow.

Study / report	**System / source identity**	**QA**
S001 / P001	VastPipe (2025)	13/18
S002 / P002	iGniter (2023)	18/20
S003 / P003	gpulets (2022)	20/20
S004 / P004	PipeSwitch (2020)	17/20
S005 / P005	REEF conference (2022)	18/20
S006 / P006	Paella: Low-latency Model Serving with Software-defined GPU Scheduling (2023)	16/18
S007 / P007	Orion: Interference-aware, Fine-grained GPU Sharing for ML Applications (2024)	19/20
S008 / P008	GSLICE: Controlled Spatial Sharing of GPUs for a Scalable Inference Platform (2020)	17/20

Study / report	System / source identity	QA
S009 / P009	Serving DNNs like Clockwork: Performance Predictability from the Bottom Up (2020)	18/20
S010 / P010	Colocating ML Inference and Training with Fast GPU Memory Handover (SIRIUS) (2025)	17/20
S011 / P011	ParvaGPU: Efficient Spatial GPU Sharing for Large-Scale DNN Inference in Cloud Environments (2024)	14/20
S012 / P012	Torpor: GPU-Enabled Serverless Computing for Low-Latency, Resource-Efficient Inference (2025)	15/20
S013 / P013	StreamBox: A Lightweight GPU SandBox for Serverless Inference Workflow (2024)	17/20
S014 / P014	Efficient Performance-Aware GPU Sharing with Compatibility and Isolation through Kernel Space Interception (KRYPTON) (2025)	16/20
S015 / P015	XSched: Preemptive Scheduling for Diverse XPUs (2025)	18/20
S016 / P018	InferFair: Towards QoS-aware scheduling for performance isolation guarantee in heterogeneous model serving systems (2024)	17/20
S017 / P019	ArkGPU: enabling applications' high-goodput co-location execution on multitasking GPUs (2023)	16/20
S018 / P020	Dyna-P: placement-aware dynamic partitioning for lightweight applications with modern GPUs (2025)	15/20
S019 / P021	KRISP: Enabling Kernel-wise RIght-sizing for Spatial Partitioned GPU Inference Servers (2023)	18/20
S020 / P022	Tally: Non-Intrusive Performance Isolation for Concurrent Deep Learning Workloads (2025)	19/20
S021 / P024	LithOS (2025)	17/20
S022 / P025	Abacus (2021)	17/20
S023 / P026	Miriam (2023)	16/20
S024 / P027	SGDRC (2025)	16/20
S025 / P028	Pantheon (2024)	16/20
S026 / P029	Oneiros (2025)	17/20
S027 / P030	GACER (2024)	13/18

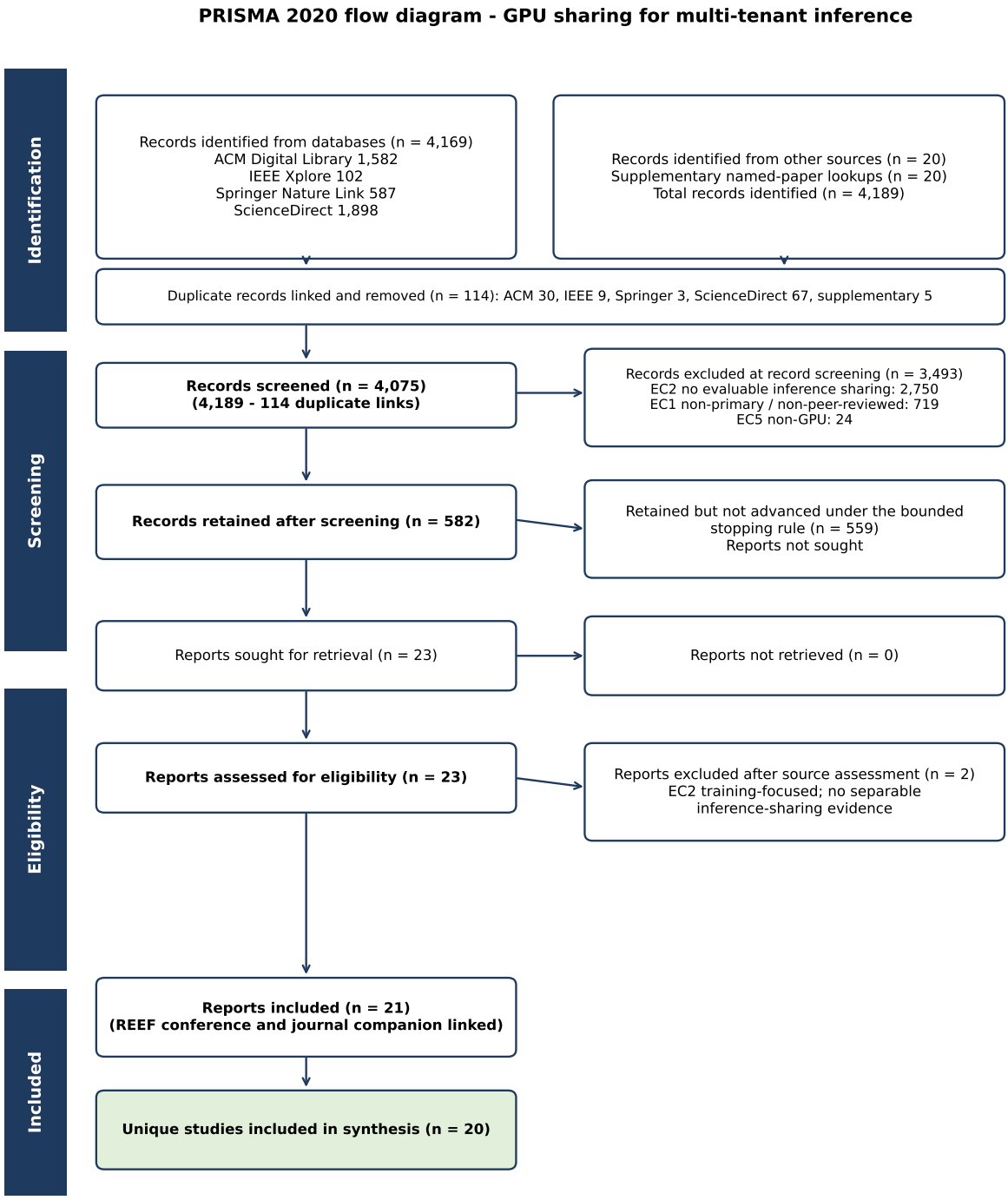


Figure 1: Frozen Task 2 selection flow, shown as historical provenance only. The adjacent branch table records later decisions; a reconciled final whole-review PRISMA flow remains pending.

5 Taxonomy of existing approaches

A useful taxonomy separates the resource being controlled from the software layer implementing the policy. The same system may appear in more than one category. A MIG-based system can also use MPS; an API interceptor can enforce temporal scheduling; and memory handover can support mixed-workload co-location. Treating these as mutually exclusive labels would hide the source of the observed behavior.

5.1 Temporal and priority control

Temporal mechanisms decide which work runs and when another tenant can make progress. PipeSwitch emphasizes switching and model state, Clockwork emphasizes predictable execution and admission, while REEF, Paella, Orion and XSched expose different scheduling/preemption boundaries. Their integration costs and latency meanings differ. PipeSwitch's additional switching-related latency is not interchangeable with XSched's foreground P99 ratio or Paella's reported P99 reduction [17, 15, 18, 19, 20, 2].

LithOS extends temporal control with transparent kernel atomization and TPC allocation; Abacus makes operator overlap deterministic so a controller can predict concurrent-service latency. Miriam and Pantheon target edge inference, but at different intervention boundaries: elastic kernels versus chunk preemption combined with early-exit variants. They are not interchangeable implementations of the same scheduler [9, 10, 11, 13].

5.2 Spatial allocation and partitions

MPS-based allocation and software right-sizing regulate concurrent access to execution resources. VastPipe, iGniter, gpulets and GSLICE provide different allocation, profiling and feedback approaches. KRISP contributes AMD-oriented kernel right-sizing evidence and should not be relabeled as a measurement of NVIDIA MPS. Partition-aware designs, including ParvaGPU and Dyna-P, additionally make placement, profile size or reconfiguration relevant [21, 22, 23, 24, 1, 25, 26].

A spatial allocation is only one part of a service guarantee. The model must fit memory, the workload must remain within the profiled regime and the selected resource shares must satisfy its timing requirements. Hardware allocation boundaries do not by themselves establish all possible security or fault guarantees in the reviewed experiments.

5.3 API, runtime and memory mediation

Torpor, StreamBox, KRYPTON, ArkGPU and Tally intervene at different points between the application and accelerator. Their common advantage is control over resource use without necessarily changing the model's high-level purpose. Their differences include intercepted interfaces, compatibility, runtime sharing, host work and the degree to which tenants share memory or trust assumptions. SIRIUS contributes a specific memory-handover mechanism for co-located inference and training [27, 28, 29, 30, 31, 32]. SGDRC dynamically controls compute and VRAM-bandwidth contention through software mechanisms, while Oneiros remaps model-parameter memory and overlaps CPU/GPU transfers to relieve

multi-model memory pressure. Oneiros evaluates temporal and spatial configurations; its memory-management benefit is not a new fault-containment guarantee. SGDRC explicitly reports that its shared CUDA context cannot isolate DNN GPU-runtime failures [12, 14].

Family / reports	**Control and condition**	**Cost**
Temporal / priority; P004 P005 P006 P007 P009 P015 P022	Request, model, kernel or launch order. A suitable interception/preemption boundary exists	Switch cost, profiling and background slowdown
Software spatial sharing; P001 P002 P003 P008 P021	MPS quota or AMD CU-mask control. Concurrent jobs fit memory and selected compute allocation	Residual interference; quotas are not proof of security
Hardware partition and hybrid; P011 P014 P020	MIG profiles plus temporal/MPS policy. Supported partition profiles fit model memory	Fragmentation, placement and reconfiguration cost
API / runtime mediation; P006 P012 P013 P014 P019 P022	User API, shared runtime or kernel interface. Required APIs and integration hooks are supported	Compatibility, CPU/transfer cost and trust boundary
Memory handover; P010	Inference/training memory ownership. Training can yield memory without breaking inference	Reclamation and integration overhead
Fairness policy; P018	Relative model slowdown and SLO goodput. Comparable per-model service objectives	Not equivalent to request-tail or hostile-tenant isolation
Fine-grained temporal / spatial control; P024 P025 P026	TPC/atomization, deterministic operator overlap and elastic kernels; compatibility and profiled workload assumptions	Kernel support, profiling, latency versus throughput
Software compute/bandwidth control; P027	SM allocation and software VRAM-bandwidth control; shared CUDA context	No DNN runtime-fault isolation
Model-adaptive edge preemption; P028	Chunk preemption plus early-exit variants; permitted accuracy/memory budget	Deadline behavior changes both model work and scheduler cost
Parameter-memory virtualization; P029	Temporal/spatial multi-model service with asynchronous transfer; high CPU/GPU bandwidth	Transfer overhead, memory policy and isolation claims separate

Family / reports	Control and condition	Cost

Cross-stack extension. GACER belongs to compiler/runtime coordination: spatial microbatch splitting and temporal cross-model scheduling complement native streams. This resource-efficiency control does not assert a fault/security boundary [33].

6 Results and evidence synthesis by research question

6.1 RQ1: mechanisms and control boundaries

The corpus supports a layered answer rather than a product list. Temporal control manages ordering and interruption; software spatial control manages concurrent allocation; hardware partitioning constrains resource placement; and API/runtime mediation supplies an enforcement layer. Fairness-aware policy adds a decision objective across these mechanisms. InferFair, for example, is best understood through relative slowdown and useful service, not as a new hardware partition primitive [34].

This taxonomy explains why naming MPS, MIG or remoting is insufficient for engineering selection. A deployment also needs a policy, a mechanism for enforcing it and an explicit boundary of protection. The corpus demonstrates several combinations, but its bounded selection does not support a claim that one category is most prevalent in the entire literature.

The seven additions reinforce the layered taxonomy. LithOS coordinates TPC scheduling and atomization, Abacus coordinates operator overlap, Miriam transforms kernels, Pantheon combines chunk preemption with model adaptation, SGDRC controls compute and memory bandwidth, and Oneiros controls parameter-memory residency. The enforcement point and shared state explain which mechanisms can be composed and which integration assumptions must be checked [9, 10, 11, 12, 13, 14].

6.2 RQ2: measurements and comparability

The outcome register includes requests/s, normalized throughput, goodput, P99/P98 latency, unspecified latency, SLO-violation rate, API overhead, host memory and a composite utilization-efficacy score. These measure different properties. Goodput counts work that meets a service objective, while aggregate throughput may include best-effort work that does not protect the foreground service. A numerical increase in the latter cannot establish improvement in the former.

GSLICE's GUE value of 94.04 is a composite score, not 94.04% GPU utilization. Its negative signed latency improvement is retained as an unfavorable observation. The gpulets threshold covering 90% of tested model pairs concerns a distribution across pairs, not the 90th percentile of requests. These distinctions prevent apparently precise but invalid conclusions [23, 24].

The reported iGniter prediction-error range of 1.53–5.02% assesses prediction accuracy, not a latency speedup. Clockwork's approximately 800 requests/s is goodput in a particular evaluation, and the supplied extraction notes a nonidentical comparator setup. Both

observations are useful, but neither supports a direct ranking against a throughput ratio measured on another platform [22, 20].

Pantheon's deadline-adjusted relative accuracy treats a late job as zero accuracy; it is not ordinary classification accuracy. Abacus excludes deadline-dropped queries from latency calculations while including them in QoS violations, so latency must be read with the violation population. Oneiros separates P99 time-to-first-token (TTFT), including queuing, from time-between-token (TBT), and reports throughput in generated tokens/s. None is directly interchangeable with DNN requests/s or an unspecified latency statistic [10, 13, 14].

6.3 RQ3: conditions affecting results

Three conditions recur in the evidence. First, concurrency and resource shares affect both completed work and waiting time. The local ParvaGPU contrast shows a throughput benefit accompanied by increased latency. Second, model state and memory transfer can dominate. Torpor's late binding is attractive for low-rate functions, but its density result does not establish that frequent transfers of large models meet the same latency objective. Third, the workload's priority structure determines which sacrifice is acceptable [1, 28].

For a foreground-priority service, XSched's 1.07x standalone P99 is relevant because it directly expresses foreground interference in the evaluated setup. Orion's up-to-7.3x aggregate-throughput result describes total work and must be read with its foreground-latency condition. These results can support different decisions without contradicting each other [19, 2].

Mixed-workload evidence requires an additional separation. SIRIUS reports an average P99 inference improvement factor of 12.6 against the authors' adapted TaskSwitch baseline. It is not a uniform per-model gain or a direct comparison against unmodified PipeSwitch. ArkGPU's recorded inference/training pairs meet a twice-solo P99 target, while a low-contention setting can lose about 5% QPS. The unfavorable condition belongs in the synthesis because it changes when the mechanism is worthwhile [27, 31].

Miriam's RTX2060 and edge-device results differ by workload and arrival mode; the introductory latency-overhead summary is not a universal bound for the body experiments. Pantheon reports 0.39–1.10% deadline misses for three stated application mixes, but does so with model-variant, memory and accuracy constraints. Oneiros's C1 spatial-sharing comparison reports author-averaged reductions of 65.5% in P99 TBT and 20.7% in P99 TTFT, with 6.6% higher tokens/s, under the stated trace and model mix. These are within-source conditions, not transferable service guarantees [11, 13, 14].

6.4 RQ4: isolation and implementation cost

Performance isolation asks whether another workload changes service behavior. Resource isolation asks what capacity is allocated or separated. Memory protection and fault containment concern different failure or adversarial models. A scheduler that limits slowdown is not thereby a complete hostile-tenant security boundary. The extraction cards preserve these dimensions separately, and untested boundaries remain unestablished.

Implementation costs also operate at different levels. API mediation can add host-side work or per-call delay. Model swapping adds transfer and memory pressure. Kernel-space

interception introduces compatibility and maintenance requirements. Temporal granularity changes the trade-off between scheduling work and waiting time. The KRYPTON extraction reports an average QoS violation rate of 1.23% relative to a solo P99 threshold, which is a threshold-relative request fraction rather than a latency in milliseconds. Its original Poisson rate is half the solo maximum; the shared test halves that rate again while assigning 50% time [30].

SGDRC's shared-context execution provides performance control without DNN runtime-fault isolation. Pantheon prioritizes deadlines by trading some model accuracy and memory residency for schedulability, and adds device power relative to its comparator. Abacus incurs profiling and predictor work. LithOS requires compatible interception and kernel behavior; Oneiros relies on high CPU/GPU transfer bandwidth and must cap remapping to avoid extra latency. These costs should be compared to the deployment's constraint, not collapsed into a single isolation or overhead score [9, 10, 12, 13, 14].

6.5 RQ5: conditional selection

The evidence supports a sequence of decisions. Specify the service objective and trust boundary first. Eliminate mechanisms incompatible with hardware, model memory or required integration. Compare the remaining candidates under the relevant arrival process, batch and co-tenant mix. Measure both protected service and work sacrificed. If the available studies do not cover the required boundary, the appropriate conclusion is insufficient evidence for that guarantee, not an assumed benefit.

The recommendation matrix in Section 10 connects each scenario to report IDs, expected benefits, costs and alternatives. These are evidence-supported candidates for evaluation, not claims that the reviewed implementation can be deployed unchanged in every environment.

The added evidence broadens the candidate set without changing the selection rule. For predictable co-located DNN service, deterministic overlap or fine-grained priority control is a candidate when profiling and compatible kernels are acceptable. For an edge deadline objective, model-changing approaches require an explicit accuracy budget. For memory-constrained multi-LLM service on a high-bandwidth CPU/GPU system, parameter remapping is a candidate while transfer cost remains tolerable. Fault-sensitive or mutually untrusted tenants require a separately established boundary [9, 10, 11, 12, 13, 14].

7 Comparative analysis

The comparison distinguishes direct within-source contrasts from unmatched cross-study observations. The strongest numerical statements are tied to their original comparator. Even these require attention to whether the result is an average, maximum, lower bound or threshold-relative statistic. An improvement described as "up to" is not the expected result for an arbitrary deployment.

Evidence / class	Observation	Interpretation and missing control
CMP01 P011 E036 E037 C001 C002. Within-source trade-off	354 to 446 requests/s; 11 to 27 ms	25.99% throughput increase accompanies 2.4545x latency. Missing: Latency quantile; final version equivalence
CMP02 P001 E001; P011 E036. Unmatched cross-study	2.7x versus 25.99% increase	Different baseline and GPU/model context prevent ranking. Missing: Matched workload, batch and resource budget
CMP03 P007 E029 E030; P015 E044. Different objectives	Up to 7.3x aggregate work versus 1.07x foreground P99	Foreground delay and total completed work must be assessed together. Missing: Common model mix and background progress metric
CMP04 P008 E031 E032. Within-source metric tension	GUE 94.04 and negative signed latency improvement	Composite efficacy is not utilization percent; preserve adverse latency. Missing: Metric definitions and baseline stratum
CMP05 P012 E038 E039. Condition-dependent trade-off	Density benefit with model-transfer sensitivity	Late binding suits low-rate functions only if transfer fits SLO. Missing: Heavy-model transfer and host CPU budget
CMP06 P019 E048 E049; P022 E052. Different experiments, consistent cost	ArkGPU approximately 5%low-contention QPS regression; Tally total normalized throughput1.19 versusTGS1.65	Foreground protection can carry aggregate or co-tenant costs; Tally outcome is total system throughput.. Missing: Matched slowdown definitions
CMP07 P005 P023. Companion study, not replication	REEF engines and report versions differ	Retain source-version strata; do not double-count corroboration. Missing: No current combined numerical rows: E018-E026 explicitly assigned to conferenceP005; journal experiments remain separate.
CMP08 P018 E046 E047; P020 E050. Different metrics	Goodput/fairness versus makespan	Job-level efficiency cannot establish request-tail service. Missing: Request arrival process and latency distribution

Evidence / class	Observation	Interpretation and missing control
CMP09 P024 / LITH-01–02. Within-source tenant-set limitation	LithOS SLO attainment100%; normalized throughput1.0 versus MPS1.11 in the logged scenario	MIG/Limits omit BE while LithOS/MPS include it; no matched all-family ranking
CMP10 P025 / ABAC rows. Population distinction	p99 improvement and peak query throughput use distinct load/settings	Deadline-dropped queries excluded from latency but counted in violations
CMP11 P026 P028. Unmatched edge experiments	Elastic kernels versus model-variant/chunk preemption	Different models, platforms and accuracy semantics; not a direct winner comparison
CMP12 P027 / SGDRC rows. Goodput versus raw throughput	Latency-sensitive protection with BE progress; adverse raw throughput retained	Shared context lacks runtime-fault isolation; baseline reimplementation matters
CMP13 P029 / ONEI-01–03. Within-source spatial case	65.5% lower P99 TBT; 20.7% lower P99 TTFT; 6.6% higher tokens/s	C1 model mix and traces; source averages, not a general hardware effect

7.1 Compiler/runtime coordination and batch-latency evidence

GACER extends the fine-grained compiler/runtime family through microbatch splitting and staged cross-model operator scheduling over GPU streams. Its final ICCAD2024 report evaluates heterogeneous inference models on one A100 and separately on P6000 and GTX1080Ti devices. Table2 reports3.30ms for VGG16+LSTM versus4.32ms sequential and22.14ms for five models versus37.86ms sequential. These are batch-completion measurements, not per-request tail latencies or online SLO attainment. Achieved SM occupancy is likewise distinct from GPU busy-time utilization. Table3 reports0.84–1.26seconds of optimization across the stated mixes and batches. This supports a conditional choice for known multi-model batch workloads when optimization and transformation costs are tolerable; it does not establish fault isolation or superiority on online tail-SLO objectives. The final report has four authors and nine pages, differing from the older seven-author preprint, which does not supply the extracted final results [33].

The evidence tables in Appendices C and H retain 52 inherited and 14 new observations, respectively. The comparative matrix is deliberately smaller because its role is to explain a decision, not duplicate the extraction register. A claim can be traced from a recommendation to a report, then to an outcome ID and its page/figure anchor. Where a comparison remains unmatched, the matrix names the missing control rather than calculating a misleading combined effect.

8 Engineering trade-off analysis

8.1 Utilization versus latency

Additional concurrency is beneficial only while useful service increases under the required latency objective. The ParvaGPU example changes throughput by

$$(446/354 - 1) \times 100 = 25.99\%,$$

while latency changes by

$$27/11 = 2.4545.$$

These are calculations from recorded inputs, not newly measured outcomes. A service with a tight delay budget might reject this configuration despite its higher throughput. Another service might accept it if the increased latency remains within its own budget. Because the source quantile is unspecified, neither interpretation establishes P99 compliance [1].

8.2 Foreground protection versus background progress

Priority is an allocation choice. Tally's supplied contrast records normalized total system throughput of 1.19 versus 1.65 for TGS at the 10% idle-time setting. This does not make foreground protection ineffective; it identifies the work sacrificed. Similarly, an aggregate-throughput gain must not hide deterioration in the protected request class. The engineering evaluation should report foreground latency and background progress together [32].

8.3 Memory density versus transfer and trust costs

Shared runtimes and late binding can allow more functions or models to use limited GPU memory. StreamBox's workflow-oriented sharing and Torpor's binding decisions address this problem from different directions. Their density gains cannot be separated from concurrency, model-loading and trust assumptions. When large models are invoked frequently, transfer demand can undermine the benefit of keeping fewer models resident. When functions require stronger separation, a shared runtime may not provide the required boundary. Torpor explicitly distinguishes runtime-sharing evaluation from runtime-isolation deployment, and describes restart/migration recovery; these design claims are not direct fault-injection results [28, 29].

8.4 Deadlines versus model accuracy and memory

Pantheon gives a different trade-off from a scheduler that preserves the original model unchanged: its early-exit variants reduce work when needed to protect deadlines. The reported deadline-adjusted accuracy therefore couples model quality and timeliness. Its similar utilization to RTm-DL does not establish universal utilization superiority. Miriam likewise requires workload-specific reading of throughput and critical-task latency rather than importing its introductory summary as a hard bound [11, 13].

8.5 Memory capacity versus transfer bandwidth

Oneiros trades parameter residency for KV-cache capacity and asynchronous transfer. Its spatial-sharing gains occur in a stated GH200 setup and model mix; more aggressive remapping can increase average latency, and halting remapping at low load can help. This complements, rather than directly supersedes, Torpor's low-rate late-binding evidence because model scale, transfer path and metrics differ [28, 14].

8.6 Partition control versus flexibility

Hardware profiles constrain placement and memory fit. Software policies can improve use of a chosen profile, but fine-grained application demand need not align with fixed partitions. Dyna-P's recorded sensitivity to prediction perturbation concerns makespan, while online serving additionally needs evidence about individual request delays. The review therefore treats placement efficiency as relevant but insufficient evidence for latency-sensitive request serving [25].

9 Discussion and conflicting evidence

9.1 Apparent conflicts versus controlled disagreements

A valid conflicting-evidence analysis first asks whether two studies measure the same quantity under comparable conditions. VastPipe's 2.7x throughput contrast and the calculated 25.99% ParvaGPU increase have different baselines and workloads. Their unequal magnitudes do not show disagreement. Likewise, Orion's aggregate throughput, SIRIUS's average P99 improvement and XSched's normalized foreground P99 do not compete on a common scale [21, 19, 27, 1, 2].

More useful tensions occur inside an experiment: improved throughput accompanied by higher latency, or protection of foreground work accompanied by lower background progress. The negative GSLICE latency contrast and ArkGPU's low-contention regression show conditions in which a favorable overall mechanism narrative is incomplete. These observations constrain applicability rather than being dismissed as exceptions [24, 31].

The added papers supply further within-source tensions. LithOS's plotted SLO advantage must be read with an unequal supported tenant set for some comparators. SGDRC can trade raw aggregate throughput for latency-sensitive goodput; it does not dominate every throughput baseline. Pantheon's lower deadline-miss rate coexists with model/memory changes, and its power experiment reports 10.15 W versus RTm-DL's 9.73 W. These are conditional costs, not mutually contradictory claims across unmatched systems [9, 12, 13].

9.2 Version and engine effects

The REEF journal extension adds implementation and evaluation beyond the conference report. The supplied extraction already distinguishes PyTorch and TensorRT conditions in its discussion. Combining all favorable points into one synthetic result would obscure both the engine difference and the report-version difference. All nine retained REEF rows

E018–E026 are now explicitly assigned to the conference P005 and its AMD evaluation stratum. P023 supplies companion context; its expanded NVIDIA and LLM experiments are separate and are not transplanted into those rows [15, 16].

9.3 Strength of the engineering argument

The repeated pattern is that resource sharing works through a mechanism-specific constraint: interruption cost, quota fit, memory residency, partition placement or compatibility. This pattern supports conditional design reasoning. It does not establish identical effect sizes or a common security guarantee across systems. The draft's recommendations remain narrower than a claim of deployment readiness.

10 Evidence-based engineering recommendations

REC01: Trusted resident models; utilization priority

Dimension	Assessment
Candidate	Tuned MPS / software spatial controls
Evidence	P001 P002 P003 P008 E001-E004 E006 E031-E032
Expected benefit	Higher useful co-location under tested allocations
Sacrifice	Interference, quota tuning and memory pressure
Alternative / trigger	Partitions or dedication when trust/isolation dominates
Validation needed	Measure each model P99 and SLO goodput under bursts

REC02: Foreground tail-latency priority

Dimension	Assessment
Candidate	Priority/preemptive scheduling
Evidence	P005 P006 P007 P015 P022 E018-E030 E044-E045 E052
Expected benefit	Lower foreground interference
Sacrifice	Background slowdown, profiling and runtime integration
Alternative / trigger	Dedicated GPU when supported control cannot meet deadline
Validation needed	Check foreground tails and background progress jointly

REC03: Low-rate serverless functions

Dimension	Assessment
Candidate	Late binding / controlled runtime sharing
Evidence	P012 P013 E038-E041
Expected benefit	Higher function or workflow density
Sacrifice	Host CPU, memory transfer and shared-runtime trust
Alternative / trigger	Keep large or high-rate models resident if swaps dominate
Validation needed	Sweep model sizes, cold starts, offered rate and concurrency

REC04: Mixed inference with opportunistic training

Dimension	Assessment
Candidate	Memory handover / controlled co-location
Evidence	P010 P019 E035 E048-E049
Expected benefit	Protect inference while using residual capacity
Sacrifice	Memory/reclamation costs and possible co-tenant regression; ArkGPU approximately 5%QPS loss under low contention.
Alternative / trigger	Separate resources for intolerable interference
Validation needed	Measure inference tails separately from training throughput

REC05: Supported partition hardware; stable memory fit

Dimension	Assessment
Candidate	MIG-aware hybrid policy
Evidence	P011 P014 P020 E036-E037 E042-E043 E050
Expected benefit	Resource allocation with configurable co-location
Sacrifice	Fragmentation and reconfiguration overhead
Alternative / trigger	Software sharing when partition granularity wastes capacity

Dimension	Assessment
Validation needed	Validate profile fit and online request tails; Dyna-P8.09%makespan penalty at15%prediction perturbation is local, not Section4.3.2simulation.

REC06: Several models with unequal service pressure

Dimension	Assessment
Candidate	Fairness-aware admission / scheduling
Evidence	P018 E046-E047
Expected benefit	In `Table3ratio3:1:2,total600requests/s:` goodput524.29 versusClockwork457.24; unfairness1.05 versus1.50.
Sacrifice	May give up aggregate throughput to improve fairness
Alternative / trigger	Priority scheduling when one foreground SLA dominates
Validation needed	Validate intended SLO/arrival mix; source uses100msSLO,50msperiod and max/min slowdown unfairness. No universal throughput advantage is inferred.

REC07: Predictable multi-model DNN co-location

Abacus and LithOS broaden the candidate set where deterministic overlap or fine-grained resource control is compatible with the workload. Evaluate inference-only results separately from mixed training, preserve co-tenant sets, and measure profiling, host and kernel-support costs. Use a more conservative allocation when the protected service cannot tolerate the measured tail or unsupported kernels [9, 10].

REC08: Edge deadlines with an explicit accuracy budget

Consider Miriam for elastic-kernel coordination and Pantheon when chunk preemption and early-exit model variants are acceptable. The expected benefit is lower deadline interference under the evaluated mix; the sacrifice can include altered accuracy, extra resident variants, scheduling work and power. Prefer a model-preserving alternative when early exits violate task-quality requirements. Test nonperiodic arrivals, background progress and the actual platform rather than importing a single reported percentage [11, 13].

REC09: Compute/bandwidth contention under a shared trust context

SGDRC is a candidate for controlling compute and VRAM-bandwidth contention when its generated-kernel environment and shared CUDA context are acceptable. The expected

benefit is latency-sensitive goodput with best-effort progress. It cannot be selected as a DNN runtime-fault boundary: that isolation is explicitly absent. Preserve cases where another baseline has greater raw throughput [12].

REC10: Memory-constrained multi-LLM serving

Oneiros is a candidate when a high-bandwidth CPU/GPU transfer path makes parameter remapping feasible. Its stated spatial-sharing case reports lower P99TBT/TTFT and higher tokens/s than vLLM. The sacrifice is transfer and remapping overhead; aggressive remapping can worsen average delay. Retain resident parameters or reduce remapping at low load or when transfers dominate, and establish required protection separately [14].

Known heterogeneous batch mixes. Consider GACER when batch-completion efficiency is the objective and profiling/optimization delay is acceptable. Its reported0.84–1.26second optimization cost and absent serving-tail/fault evidence limit transfer to online SLO-sensitive or mutually untrusted tenants [33]. The recommended deployment assessment begins with the actual service objective: per-model deadline or percentile target, admissible violation rate and minimum background progress. These values must come from the intended deployment rather than being invented from a paper. The next step is a local comparison using the same model versions, input sizes, batching and offered load. Cold starts and bursts should be retained if they occur in the application.

A candidate is suitable only if it meets the required service objective at an acceptable cost. The acceptance evidence should include per-tenant latency, SLO-compliant goodput, memory pressure, resource utilization with a defined instrument, host overhead and the relevant isolation tests. The review supplies a shortlist and the reasons to test it; it does not supply a universal numerical threshold or imply that a benchmark result transfers unchanged.

11 Research gaps and future directions

Corpus gap	Basis	Next experiment
No matched all-family comparison	Comparability CMP01-CMP08	Same GPU, model mix, memory budget, arrivals and SLO across families
Security/fault boundaries often unestablished	Extraction Security boundary / Fault boundary	Separate adversarial access, fault injection and performance tests
Metrics emphasize different units	E001-E052	Report per-model goodput, tails and background progress jointly
Partition prediction affects makespan	P020 E050	Bursty arrivals plus placement changes and request-tail measurement

Corpus gap	Basis	Next experiment
Version and method details remain incomplete	ParvaGPU final-text equivalence; source Q3/Q4/Q10 reporting limits. iGniter final checked; Tallyv3 pinned.	Publish exact version, driver/runtime, warmup, repetitions and uncertainty
Energy and cost evidence not harmonized	Extraction schema and heterogeneous outcomes	Measure GPU and host energy per SLO-compliant request
Model-adaptive versus model-preserving schedules	P028 changes exit/variant; P026 transforms kernels	Hold accuracy budget and input distribution constant; jointly report deadline and task quality
Multi-model memory remapping and protection	P029 capacity/latency results; no fault guarantee	Measure transfer, tenant progress, tail latency and fault behavior with explicit isolation boundaries
Source configuration/report inconsistencies	P025 A100 SM label; P026 platform label; P029 platform/isolation caption	Publish exact device identifiers and corrected experiment manifests

These gaps are observations about this corpus and its reporting, not assertions that no relevant research exists anywhere. The 326 unresolved advanced records make that distinction essential. A stronger future comparison would align hardware, workload, baselines and outcome definitions across mechanism families, then report both benefits and sacrifices. Energy per useful request and failure/security behavior require separate measurements rather than extrapolation from utilization.

12 Threats to validity and limitations

Threat / consequence	Mitigation	Remaining work
Incomplete eligibility: No exhaustive recall or prevalence claim	559screening decisions preserved;56borderline proposals reopened	Close325advanced records and report identities; reconcile whole-review flow

Threat / consequence	Mitigation	Remaining work
Title-only decisions: Relevant work may be missed	Abstract-availability column retained	Prioritize uncertain/relevant records for abstract checks
Single review: Correlated screening/QA/extraction errors possible	Original rationale and source anchors retained	Record real named checks if performed
Search provenance: Search cannot be fully reconstructed from totals	Exact preserved query/log values retained	Supply missing original exports, facets and receipts
Source version: Page/result mappings may differ	iGniter final checked; Tallyv3 pinned; REEF conference/journal mapping resolved	Confirm ParvaGPU final equivalence or retain version condition.
Measurement heterogeneity: Cross-study speedup ranking is invalid	No pooled effect size or universal winner	Matched comparisons needed
Reporting bias: Headlines may overstate robustness	52 inherited plus18 new source-anchored outcomes; adverse findings retained	Independent high-impact claim checks; reporting uncertainty still missing in some sources.
New-source reporting discrepancies	Primary labels and adverse contexts preserved in appraisal/source cautions	Clarify hardware, version and isolation-caption inconsistencies; do not silently repair source claims

The most material limitation is selection: a bounded source set cannot establish exhaustive coverage. The large number of title-only records compounds that risk. The single-assessment method creates a separate risk of correlated errors even where spreadsheet arithmetic is correct. Finally, retrieved author manuscripts may differ from the version used by the original extraction, and access to a PDF does not constitute verification of every result.

Mitigations in this draft include immutable baseline files, separate current retrieval records, stable IDs, preserved unfavorable findings, explicit source/version limits and no numerical pooling across unmatched studies. These measures improve traceability but do not erase the underlying limitations. Final acceptance requires resolving the scope decision and recording any actual independent checks at the item level.

13 Conclusion

The working 27-study corpus supports a conditional approach to GPU sharing for multi-tenant inference. Temporal control is relevant when foreground response dominates; software spatial sharing is relevant when trusted co-location can exploit spare capacity; partition-aware designs make resource fit and reconfiguration central; and runtime or memory mediation can increase density at a transfer, integration or trust cost. Each choice must be evaluated against its own baseline and intended conditions.

The principal outcome is a traceable selection framework. The 13 main report sections now integrate the seven confirmed additions into taxonomy, metric interpretation, trade-offs and conditional recommendations. The work remains an evolving draft: 325 advanced records await eligibility or identity closure, whole-review PRISMA requires reconciliation, and historical search and source-version limitations remain explicit. Further eligible studies must be integrated using the same criteria, rather than stopping at the current count.

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[28] M. Yu, A. Wang, D. Chen, H. Yu, X. Luo, Z. Li, W. Wang, R. Chen, D. Nie, H. Yang, and Y. Ding, "Torpor: GPU-Enabled Serverless Computing for Low-Latency, Resource-Efficient Inference," in *2025 USENIX Annual Technical Conference (ATC)*, pp. 597–612, 2025. Report P012, study S012. Runtime-sharing evaluation and runtime-isolation deployment are distinguished.

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A Exact search register

The following entries are preserved from the supplied workbook. Date precision, provenance and count discrepancies remain as recorded.

A.1 IEEE-MPS

Field	Recorded value
Platform	IEEE Xplore
Status	executed search record preserved in the project register
Exact query	(GPU OR "graphics processing unit") AND (inference OR "model serving" OR "deep learning" OR DNN OR LLM) AND (MPS OR "multi-process service")
Fields	All metadata (Search by All); not field-qualified Command Search
Filters	Publication years2020-2026; date/language/type checked during screening
Recorded search date / time	2026-10-11
Displayed hits	25
Exported records	25
Parsed records	25
Active records	25
Files	https://github.com/Raazia-Imran/os_CEPPPPPP/blob/main/evidence/imports/group-a/imported_records.json
Notes	Rendered metadata capture, not publisher export; exact time unavailable; different expressions from submitted protocol; full text indexed scope for Springer
Provenance	Group A run, reused (counts identical to Group A report, 16 Oct 2026)

A.2 IEEE-MIG

Field	Recorded value
Platform	IEEE Xplore

Field	Recorded value
Status	executed search record preserved in the project register
Exact query	(GPU OR "graphics processing unit") AND (inference OR "model serving" OR "deep learning" OR DNN OR LLM) AND (MIG OR "multi-instance GPU" OR "GPU partitioning")
Fields	All metadata (Search by All); not field-qualified Command Search
Filters	Publication years2020-2026; date/language/type checked during screening
Recorded search date / time	2026-10-11
Displayed hits	23
Exported records	23
Parsed records	23
Active records	23
Files	https://github.com/Raazia-Imran/os_CEPPPPPP/blob/main/evidence/imports/group-a/imported_records.json
Notes	Rendered metadata capture, not publisher export; exact time unavailable; different expressions from submitted protocol; full text indexed scope for Springer
Provenance	Group A run, reused (counts identical to Group A report, 16 Oct 2026)

A.3 IEEE-TEMPORAL

Field	Recorded value
Platform	IEEE Xplore
Status	executed search record preserved in the project register
Exact query	(GPU OR "graphics processing unit") AND (inference OR "model serving" OR "deep learning" OR DNN OR LLM) AND ("time slicing" OR "time sharing" OR "context switching")
Fields	All metadata (Search by All); not field-qualified Command Search

Field	Recorded value
Filters	Publication years2020-2026; date/language/type checked during screening
Recorded search date / time	2026-10-11
Displayed hits	45
Exported records	45
Parsed records	45
Active records	45
Files	https://github.com/Raazia-Imran/os_CEPPPPPP/blob/main/evidence/imports/group-a/imported_records.json
Notes	Rendered metadata capture, not publisher export; exact time unavailable; different expressions from submitted protocol; full text indexed scope for Springer
Provenance	Group A run, reused (counts identical to Group A report, 16 Oct 2026)

A.4 IEEE-API

Field	Recorded value
Platform	IEEE Xplore
Status	executed search record preserved in the project register
Exact query	(GPU OR "graphics processing unit") AND (inference OR "model serving" OR "deep learning" OR DNN OR LLM) AND ("API remoting" OR "CUDA remoting" OR "GPU proxy" OR "GPU virtualization")
Fields	All metadata (Search by All); not field-qualified Command Search
Filters	Publication years2020-2026; date/language/type checked during screening
Recorded search date / time	2026-10-11
Displayed hits	9
Exported records	9

Field	Recorded value
Parsed records	9
Active records	9
Files	https://github.com/Raazia-Imran/os_CEPPPPPP/blob/main/evidence/imports/group-a/imported_records.json
Notes	Rendered metadata capture, not publisher export; exact time unavailable; different expressions from submitted protocol; full text indexed scope for Springer
Provenance	Group A run, reused (counts identical to Group A report, 16 Oct 2026)

A.5 SPR-MASTER

Field	Recorded value
Platform	Springer Nature Link
Status	executed search record preserved in the project register
Exact query	(GPU OR "graphics processing unit") AND (inference OR "model serving" OR "deep learning" OR DNN OR LLM) AND ("time slicing" OR "time sharing" OR "context switching" OR MPS OR "multi-process service" OR MIG OR "multi-instance GPU" OR "GPU partitioning" OR "API remoting" OR "CUDA remoting" OR "GPU proxy" OR "GPU virtualization")
Fields	General search/full-text indexed collection
Filters	English; Research article OR Conference paper;2020-2026; manual cutoff
Recorded search date / time	2026-10-11
Displayed hits	587
Exported records	587
Parsed records	587
Active records	587
Files	https://github.com/Raazia-Imran/os_CEPPPPPP/blob/main/evidence/imports/group-a/imported_records.json

Field	Recorded value
Notes	Rendered metadata capture, not publisher export; exact time unavailable; different expressions from submitted protocol; full text indexed scope for Springer
Provenance	Group A run, reused (counts identical to Group A report, 16 Oct 2026)

A.6 ACM-PILOT

Field	Recorded value
Platform	ACM Digital Library
Status	Blocked pilot; no records imported
Exact query	GPU AND inference AND MPS
Fields	not_validated
Filters	not_applied
Recorded search date / time	2026-10-11
Displayed hits	UNKNOWN
Exported records	0
Parsed records	0
Active records	0
Files	supplied search `export/imported_records.json`
Notes	Rendered metadata capture, not publisher export; exact time unavailable; different expressions from submitted protocol; full text indexed scope for Springer
Provenance	Pilot checkpoint of 11 Oct 2026 (search blocked/unavailable; no import), as in Group A report

A.7 SD-PILOT

Field	Recorded value
Platform	ScienceDirect
Status	Blocked pilot; no records imported

Field	Recorded value
Exact query	GPU AND inference AND MPS
Fields	not_validated
Filters	not_applied
Recorded search date / time	2026-10-11
Displayed hits	UNKNOWN
Exported records	0
Parsed records	0
Active records	0
Files	supplied search `export/imported_records.json`
Notes	Rendered metadata capture, not publisher export; exact time unavailable; different expressions from submitted protocol; full text indexed scope for Springer
Provenance	Pilot checkpoint of 11 Oct 2026 (search blocked/unavailable; no import), as in Group A report

A.8 SUP-SELECTED

Field	Recorded value
Platform	Supplementary public web/venue/author lookup
Status	executed search record preserved in the project register
Exact query	Named-paper lookups and official venue/author retrieval; see `supplementary_discovery.csv`
Fields	Public metadata/full texts
Filters	Same protocol eligibility
Recorded search date / time	2026-10-11
Displayed hits	UNKNOWN
Exported records	20
Parsed records	20
Active records	20

Field	Recorded value
Files	`evidence/imports/group-a/supplementary_discovery.csv`
Notes	20 selected named-paper records in the preserved source register; this is not an exhaustive web search. Query wording is a reproducible follow-up, not a verbatim historical query.
Provenance	Shared original-collector register, retained with its source IDs; no new search claimed.

A.9 ACM-MASTER

Field	Recorded value
Platform	ACM Digital Library
Status	reported master query; master and numbered exports not fully mapped
Exact query	(GPU OR "graphics processing unit") AND (inference OR "model serving" OR "deep learning" OR DNN OR LLM) AND ("time slicing" OR "time sharing" OR "context switching" OR MPS OR "multi-process service" OR MIG OR "multi-instance GPU" OR "GPU partitioning" OR "API remoting" OR "CUDA remoting" OR "GPU proxy" OR "GPU virtualization")
Fields	All Fields
Filters	2020 - 2026
Recorded search date / time	2026-10-14 22:30 PKT (UTC+05:00; operator-reported)
Displayed hits	1600
Exported records	1603
Parsed records	1829
Active records	0
Files	`evidence/incoming_shared/acm_master/ACM_MASTER_01.bib;evidence/incoming_shared/acm_master/ACM_MASTER_02.bib`

Field	Recorded value
Notes	Separate master exports contain 1,829 overlapping occurrences / 1,387 distinct DOI-or-title keys. Reported displayed 1,600 and claimed export 1,603 do not reproduce those bytes. Active contribution is zero; do not assign the 1,582 numbered occurrences to this run without mapping evidence. Source: `evidence/incoming_shared/acm_master/reconciliation.json`.
Provenance	Historical operator-reported run; timestamp/query from `evidence/incoming_shared/search_log_2026-10-16.txt`; not independently rerun.

A.10 SD-TEMPORAL

Field	Recorded value
Platform	ScienceDirect
Status	historical operator-reported; partial export reconciliation
Exact query	(GPU OR "graphics processing unit") AND (inference OR "model serving" OR "deep learning") AND ("time slicing" OR "time sharing" OR "context switching")
Fields	Find articles with these terms
Filters	2020-2026; supplied URL explicitly includes accessTypes=openaccess (not stated in original filter text).
Recorded search date / time	2026-10-14 23:45 PKT (UTC+05:00; operator-reported)
Displayed hits	1053
Exported records	1053
Parsed records	1053
Active records	1053

Field	Recorded value
Files	`evidence/incoming_shared/sciencedirect_2026-10-16/SD_TEMPORAL_01.ris;evidence/incoming_shared/sciencedirect_2026-10-16/SD_TEMPORAL_010.ris;evidence/incoming_shared/sciencedirect_2026-10-16/SD_TEMPORAL_011.ris;evidence/incoming_shared/sciencedirect_2026-10-16/SD_TEMPORAL_02.ris;evidence/incoming_shared/sciencedirect_2026-10-16/SD_TEMPORAL_03.ris;evidence/incoming_shared/sciencedirect_2026-10-16/SD_TEMPORAL_04.ris;evidence/incoming_shared/sciencedirect_2026-10-16/SD_TEMPORAL_05.ris;evidence/incoming_shared/sciencedirect_2026-10-16/SD_TEMPORAL_06.ris;evidence/incoming_shared/sciencedirect_2026-10-16/SD_TEMPORAL_07.ris;evidence/incoming_shared/sciencedirect_2026-10-16/SD_TEMPORAL_08.ris;evidence/incoming_shared/sciencedirect_2026-10-16/SD_TEMPORAL_09.ris`
Notes	TEMPORAL URL has openaccess facet omitted from filter text; 1053 parsed agrees with reported count; exact pagination and operator unknown Repository check: standalone `evidence/incoming_shared/SD_TEMPORAL_01(1).ris` contains 25 records only; it is not the 1053-occurrence full TEMPORAL archive cited in the project register. Later archive files are not present in the inspected remote tree. Original reported/registered counts are preserved, not revalidated from this sample.
Provenance	Historical operator-reported run; timestamp/query from `evidence/incoming_shared/search_log_2026-10-16.txt`; not independently rerun.

A.11 SD-MPS

Field	Recorded value
Platform	ScienceDirect
Status	historical operator-reported; partial export reconciliation
Exact query	(GPU OR "graphics processing unit") AND (inference OR "model serving" OR "deep learning") AND (MPS OR "multi-process service")
Fields	Find articles with these terms
Filters	2020 - 2026

Field	Recorded value
Recorded search date / time	2026-10-14 23:52 PKT (UTC+05:00; operator-reported)
Displayed hits	517
Exported records	563
Parsed records	563
Active records	563
Files	`evidence/incoming_shared/sciencedirect_2026-10-16/SD_MPS_01.ris;evidence/incoming_shared/sciencedirect_2026-10-16/SD_MPS_02.ris;evidence/incoming_shared/sciencedirect_2026-10-16/SD_MPS_03.ris;evidence/incoming_shared/sciencedirect_2026-10-16/SD_MPS_04.ris;evidence/incoming_shared/sciencedirect_2026-10-16/SD_MPS_05.ris;evidence/incoming_shared/sciencedirect_2026-10-16/SSD_MPS_06.ris`
Notes	563 parsed versus 517 displayed; export claim 563; operator and exact pagination unknown Repository check: standalone `evidence/incoming_shared/SD_MPS_01.ris` contains 25 records only; it is not the 563-occurrence full MPS archive cited in the project register. Later archive files are not present in the inspected remote tree. Original reported/registered counts are preserved, not revalidated from this sample.
Provenance	Historical operator-reported run; timestamp/query from `evidence/incoming_shared/search_log_2026-10-16.txt`; not independently rerun.

A.12 SD-MIG

Field	Recorded value
Platform	ScienceDirect
Status	historical operator-reported; partial export reconciliation
Exact query	(GPU OR "graphics processing unit") AND (inference OR "model serving" OR "deep learning") AND (MIG OR "multi-instance GPU" OR "GPU partitioning")
Fields	Find articles with these terms
Filters	2020 - 2026

Field	Recorded value
Recorded search date / time	2026-10-14 23:55 PKT (UTC+05:00; operator-reported)
Displayed hits	241
Exported records	241
Parsed records	241
Active records	241
Files	`evidence/incoming_shared/sciencedirect_2026-10-16/SD_MIG_01.ris;evidence/incoming_shared/sciencedirect_2026-10-16/SD_MIG_02.ris;evidence/incoming_shared/sciencedirect_2026-10-16/SD_MIG_03.ris`
Notes	Parsed count agrees with reported count; identity and page completeness unverified; operator unknown Repository check: standalone `evidence/incoming_shared/SD_MIG_01.ris` contains 25 records only; it is not the 241-occurrence full MIG archive cited in the project register. Later archive files are not present in the inspected remote tree. Original reported/registered counts are preserved, not revalidated from this sample.
Provenance	Historical operator-reported run; timestamp/query from `evidence/incoming_shared/search_log_2026-10-16.txt`; not independently rerun.

A.13 SD-API

Field	Recorded value
Platform	ScienceDirect
Status	historical operator-reported; partial export reconciliation
Exact query	(GPU OR "graphics processing unit") AND (inference OR "model serving" OR "deep learning") AND ("API remoting" OR "CUDA remoting" OR "GPU proxy" OR "GPU virtualization")
Fields	Find articles with these terms
Filters	2020 - 2026
Recorded search date / time	2026-10-14 23:56 PKT (UTC+05:00; operator-reported)

Field	Recorded value
Displayed hits	41
Exported records	41
Parsed records	41
Active records	41
Files	`evidence/incoming_shared/sciencedirect_2026-10-16/SD_API_01.ris`
Notes	Parsed count agrees with reported count; identity and page completeness unverified; operator unknown Repository check: standalone `evidence/incoming_shared/SD_API_01.ris` contains 25 records only; it is not the 41-occurrence full API archive cited in the project register. Later archive files are not present in the inspected remote tree. Original reported/registered counts are preserved, not revalidated from this sample.
Provenance	Historical operator-reported run; timestamp/query from `evidence/incoming_shared/search_log_2026-10-16.txt`; not independently rerun.

A.14 ACM-NUMBERED-UNMAPPED

Field	Recorded value
Platform	ACM Digital Library
Status	actual supplied exports; run mapping unknown
Exact query	unknown
Fields	unknown
Filters	unknown
Recorded search date / time	unknown
Displayed hits	unknown
Exported records	unknown
Parsed records	1582
Active records	1582

Field	Recorded value
Files	evidence/imports/acm/*.bib; exact files in import_manifest.csv
Notes	79 preserved files contain 1,582 occurrences with hashes in `evidence/import_manifest.csv`. Filename number 3 is absent; this is not proof of a missing page. Exact query/date/field mapping for these files is not documented. Keep separate from the two overlapping master exports.
Provenance	Original user-supplied numbered ACM exports; repository manifest supports count and hashes, not historical run mapping.

B Original 20-study appraisal and extraction cards

The following inherited cards preserve the original source package. The seven new cards are in Appendix H; together they cover the working 27-study corpus.

These cards integrate the documented source-review corrections for all 20 included studies. Original forms remain in the frozen baseline. They are an audit appendix; the main text is organized by mechanism and RQ. Scores are not a claim of independent verification.

B.1 S001 / P001 VastPipe

[21]

Field	Extracted content
Mechanism	MPS adaptive quotas; separate AMD CU-mask stratum
Hardware and environment	NVIDIA A100 for MPS; AMD MI100 for CU-mask tests
Runtime	NVIDIA TensorRT serving; AMD TVM/ROCm with `hipExtStreamCreateWithCUMask`
Workload	CV pairs and BERT/GPT2 mixed model pairs; Poisson-generated request mixes
Tenants	Two co-located inference model services in reported local model-pair comparisons
Load and SLO	Poisson-generated mixed requests in batch-oriented co-location tests. Specific GPU-test arrival rates and latency SLO are not established in the current extract.
Comparator	No MPS; default MPS without per-service quotas; RL-selected MPS quotas. AMD no/default CU mask separately.
Finding	A100 VGG16+GoogLeNet: throughput 2.7x no MPS and 1.5x default MPS; not a universal average or p99 result. NLP/CV mean 2.43x no MPS, 1.45x default MPS. MI100 mean 1.89x no mask, 1.24x default mask.
Performance isolation	Throughput/interference comparison only; no extracted tail-latency or SLO protection measurement.
Security boundary	Not established
Fault boundary	Not tested in extracted GPU results
Overhead	Offline Q-table generation; plans reused within request windows; source Section 6.9
Limitations	Include only local GPU inference evidence, not the main 1000-NPU cluster results. Offline Q-table assumes relatively stable request mix; GPU-specific quotas, repetitions and variance unresolved. Chart bars are not preserved in Markdown. Preliminary ICCD2021 report is linked as a companion-version lead outside the synthesis corpus. GPU-specific throughput only; exact rate,batch and selected quotas remain unresolved. GPU-specific throughput only; exact rate,batch and selected quotas remain unresolved.

Field	Extracted content
Source	https://jcst.ict.ac.cn/en/article/pdf/preview/10.1007/s11390-024-3773-5.pdf
Evidence location	PDF pp16-17 / printed458-459; Figure14; section6.7

QA	Score / maximum	Rationale and anchor
Q1	2 / 2	Local GPU portability and throughput objective are explicit; NPU cluster results are excluded from this extracted stratum.. PDF pp2,16-17; section6.7
Q2	2 / 2	RL-tuned MPS allocation and MI100 CU-mask backend are described.. PDF pp16-17; sections6.7-6.8
Q3	1 / 2	GPU throughput provenance is clear, but GPU-specific repeated-run uncertainty is not established.. PDF pp16-18
Q4	1 / 2	A100/TensorRT and MI100/TVM/ROCm are named; exact versions, GPU batch/rate and tuned quotas remain incomplete.. PDF pp16-17
Q5	1 / 2	Same A100 no/default/tuned MPS comparison; default means unspecified respective quotas.. PDF pp16-17; Figure14
Q6	2 / 2	2.7x versus noMPS and1.5x versus defaultMPS are explicit for VGG16+GoogLeNet; not universal averages.. PDF p17 / printed459; Figure14 onp16
Q7	N/A	No online latency claim in the retained local GPU outcomes; NPU results cannot fill this item.. PDF pp16-17; sections6.7-6.8
Q8	2 / 2	Several paired CV/NLP model co-locations are measured.. PDF pp16-17; Figures14-16
Q9	1 / 2	Performance sharing is measured; separate security/fault validation is not established. Q9 calibration remains flagged.. PDF pp16-17
Q10	1 / 2	Stable-mix Q-table assumption and retuning for changed demand are explicit; GPU replication detail remains incomplete.. PDF p17; section6.9

B.2 S002 / P002 iGniter

[22]

Field	Extracted content
Mechanism	MPS active-thread quotas + interference-aware batch/provisioning and shadow process
Hardware and environment	Ten EC2 p3.2xlarge instances,one V100 each,8vCPU/61GB; `Triton2.12.0,TensorRT8.0.1.6.` MPS active-thread percentage; allocation unit2.5%. Heterogeneous extension adds20 T4 instances.
Runtime	NVIDIA Triton 2.12.0; TensorRT 8.0.1.6; MPS active-thread percentage (recorded hardware/setup evidence).
Workload	`AlexNet,ResNet50,VGG19(ImageNet),SSD(VOC2012);12` workload/SLO configurations; constant arrivals. Local experiment co-locates four models at25% quota,batch3. p99 provisioning measured over windows e.g.30s; model-validation experiments repeated3 times with SD bars.
Tenants	Local test co-locates four inference models at 25% MPS quota each, batch 3; separate provisioning experiments use multiple instances.
Load and SLO	batch3 for prediction check;W1 SLO10ms for recovery example
Comparator	FFD+; GSLICE+ with iGniter placement; gpu-lets+ modified batch strategy and best-fit placement. These are adapted baselines.
Finding	Local four-model latency prediction error1.53%-5.02%(Fig13). Fig14 deployment: iGniter6 GPUs/0 SLO violations vs `gpu-lets+8/3,FFD+5/10,GSLICE+6/3;`25% lower hourly instance cost vs gpu-lets+, not25% higher local utilization. Fig17 shows an initial p99 violation before shadow-process recovery.
Performance isolation	Local interference prediction and p99/SLO checks; transient prediction failure explicitly retained
Security boundary	Not established
Fault boundary	Not tested in extracted results
Overhead	Shadow standby consumes capacity/memory; extra provisioning and interference-model costs are discussed. A standalone numerical overhead is not assessed in the retained extract.

Field	Extracted content
Limitations	Constant arrivals and four vision models limit bursty/LLM transfer. Negligible PCIe contention assumption can underpredict AlexNet; baselines modified. `Steady-state/post-correction` SLO success is not an unconditional guarantee; shadow standby incurs capacity/memory costs. Chart bars/latency traces unavailable in Markdown.
Source	https://prefer-nsf.org/pdf/IEEE%20TPDS%20_%20ignite r%202023.pdf
Evidence location	Final PDF p11 / printed822; Figure13; Final PDF p12 / printed823; Figure17; section5.3

QA	Score / maximum	Rationale and anchor
Q1	2 / 2	Predictable cost-efficient inference provisioning is the stated objective.. Final PDF p1
Q2	2 / 2	MPS active-thread allocation, placement and shadow-Triton recovery are specified.. Final PDF pp8-10; section4.2
Q3	2 / 2	Model validation repeats three times with SD bars; resource plan and P99 observation windows are described.. Final PDF pp10-12; sections5.2-5.3
Q4	1 / 2	V100 instances, four models and batch/arrival settings are given; exact OS/driver coverage remains incomplete.. Final PDF p10; Table3; section5.1
Q5	1 / 2	GSLICE+ and gpulets+ are modified; batch/allocation objectives differ and modifications must remain visible.. Final PDF p10; Baselines and Metrics
Q6	2 / 2	Cost, allocation and throughput are separately defined; cluster cost is not local GPU throughput.. Final PDF pp10-13
Q7	2 / 2	P99 over e.g.30seconds and first-second15.6ms versus10ms SLO, then shadow activation at1.5seconds are explicit.. Final PDF pp11-12; section5.3; Figure17
Q8	2 / 2	Four co-located models with25% allocation and batch3 test local interference.. Final PDF p11; Figure13
Q9	2 / 2	Q9=2 for direct measured performance-interference evaluation within the stated service scope; this does not establish security or fault isolation.. PDF pp8-10; `interference/provisioning` evaluation

QA	Score / maximum	Rationale and anchor
Q10	2 / 2	Profiling cost and PCIe/prediction assumptions are disclosed; source code identified in final article.. Final PDF pp11,13-14; sections5.4,6

B.3 S003 / P003 gpulets

[23]

Field	Extracted content
Mechanism	MPS + temporal
Hardware and environment	RTX2080Ti 11GB; two/server; Ubuntu18.04; CUDA10.2; driver440.64; PyTorch1.10
Runtime	`NVIDIA440.64;CUDA10.2;PyTorch1.10` (p207 Table2)
Workload	Nine models; Poisson arrivals; per-model SLO
Tenants	paired local inferences; system scenarios contain several models
Load and SLO	Poisson inter-arrivals; model SLO5-202ms;99% requests within SLO; median of3 runs (pp207-208)
Comparator	Adapted SBP/Nexus; best-fit with/without interference
Finding	Local gpulets support interference-aware provisioning; cluster throughput is a different outcome.
Performance isolation	Interference-aware provisioning with per-model SLO constraints. Fig. 7 reports 90% of measured model pairs below 17.66% interference overhead; this is an across-pair CDF, not request p90/p99.
Security boundary	Not assessed for adversarial security in the retained source evidence; performance-sharing results do not establish memory/security isolation.
Fault boundary	Not assessed for cross-tenant fault containment in the retained source evidence; scheduling/performance results do not establish it.
Overhead	20-second reorganization interval; overlap new proxy preparation with active serving (p207 Fig11)
Limitations	Server-level gains cannot be treated as one-GPU utilization. Pairwise90%threshold describes a distribution across tested model pairs, not P90request latency. Artifact appendix provides a versioned environment. Pairwise90%threshold describes a distribution across tested model pairs, not P90request latency. Artifact appendix provides a versioned environment.
Source	https://www.usenix.org/system/files/atc22-choi-seungbeom.pdf
Evidence location	PDF p6 / printed203; Figure7; section3.4

QA	Score / maximum	Rationale and anchor
Q1	2 / 2	SLO-aware spatio-temporal sharing objective is explicit.. PDF pp2-3
Q2	2 / 2	MPS gpulet partitioning, scheduling, interference model and process replacement are described.. PDF pp7-10; sections4-5
Q3	2 / 2	Poisson arrivals and three repetitions with median result are explicit.. PDF pp10-11; section5.1
Q4	2 / 2	`CPU,RTX2080Ti,OS,driver,CUDA,PyTorch` and models are tabulated.. PDF p10; Tables2-3
Q5	2 / 2	Ported SBP and greedy best-fit conditions, and why full Nexus is not compared, are explained.. PDF p10; section5.1
Q6	2 / 2	SLO-preserved throughput is maximum rate with99% requests within SLO; multi-GPU results not substituted for local CDF.. PDF p11; section5.2
Q7	2 / 2	17.66% is the90%-of-scenarios overhead threshold, not requestP90/P99.. PDF p6 / printed203; Figure7
Q8	2 / 2	250 model/batch combinations across partition ratios test local interference.. PDF p6; section3.4
Q9	2 / 2	Q9=2 for direct measured interference and allocation-boundary evaluation; no hostile-tenant security guarantee is inferred.. PDF pp3-4,8-12; sections3 and5
Q10	2 / 2	Artifact appendix provides versioned dependencies, Docker/build/run instructions and limitations; `availability/replicability` reporting is scored without claiming a local replication. Zenodo 10.5281/zenodo.6544909 and casys-kaist/glet are supplied.. PDF pp17-18; artifact appendix: Ubuntu18.04/kernel4.15, LibTorch1.10, CUDA>=10.2, cuDNN>=7.6, Boost>=1.6, OpenCV>=4, CMake>=3.19

B.4 S004 / P004 PipeSwitch

[17]

Field	Extracted content
Mechanism	Temporal context switching
Hardware and environment	V10016GB or T416GB EC2; PyTorch1.3; CUDA10.1
Runtime	`CUDA10.1;PyTorch1.3.0;torchvision0.4.2;scipy1.3.2;`
Workload	Training batch32; inference8; RN152/InceptionV3/BERT; 100 runs
Tenants	one training task preempted for one inference task; separable inference latency
Load and SLO	`training32;inference8;100-run` averages;Fig6 reports minimum/maximum (p508)
Comparator	`Unoptimized/grouped/per-layer` transfer
Finding	Fig7: inference preemption latency reduced by up to 38.2ms.
Performance isolation	Priority inference preempts training. Table 3 reports 6.01 ms additional mean latency for ResNet152 on V100 under the recorded workload; not a tail-latency or adversarial isolation bound.
Security boundary	Not assessed for adversarial security in the retained source evidence; performance-sharing results do not establish memory/security isolation.
Fault boundary	Not assessed for cross-tenant fault containment in the retained source evidence; scheduling/performance results do not establish it.
Overhead	Table3 total extra latency;Table4 first-layer startup overhead;kept separate
Limitations	Average switch/inference latency is not p99; multi-VM throughput separate.
Source	https://www.usenix.org/system/files/osdi20-bai.pdf
Evidence location	PDF pp11-12 / printed508-509; Table3; section6.1

QA	Score / maximum	Rationale and anchor
Q1	2 / 2	Time-sharing and fast context-switching between training/inference tasks are explicit.. PDF p2; abstract
Q2	2 / 2	Pipelined transmission, controller, daemon and active/standby workers are specified.. PDF pp6-11; section4-5
Q3	2 / 2	100-run means and Figure6 min/max are explicitly reported.. PDF p11; Metrics
Q4	1 / 2	`V100/T4,PCIe,memory,PyTorch1.3.0` andCUDA10.1 given; exact OS/kernel/driver unresolved.. PDF p11; Setup
Q5	2 / 2	Ready-model baseline, same modified PyTorch and MPS `full-memory/unified-memory` swapping condition are explicit.. PDF pp11-12; section6.1
Q6	2 / 2	Utilization is throughput relative to ready-model upper bound; eight-instance scheduling-cycle result kept distinct from local table.. PDF pp12-13; Figure6
Q7	2 / 2	Table3 subtracts ready-model latency; it reports added mean overhead, not total latency or P99.. PDF pp11-12; Table3
Q8	2 / 2	Single-GPU `training-preempted-by-inference` workload has separate inference latency; qualifies under mixed-workload rule.. PDF p11; Workloads and section6.1
Q9	1 / 2	Process/memory control is described, but independent hostile-client/fault experiment is not established.. PDF pp8-11
Q10	1 / 2	Stack versions, model registration and checkpoint dependence supplied; full reproduction configuration remains incomplete.. PDF pp10-11; sections5-6

B.5 S005 / P005 REEF conference

[15]

Field	Extracted content
Mechanism	Reset-based temporal preemption + dynamic kernel padding; NVIDIA binary instrumentation
Hardware and environment	Conference: AMD MI50, 16GB, 60 CUs. Journal companion: MI50 16GB and separate NVIDIA GV100 32GB/80 SMs. Version-specific platforms are not a common configuration.
Runtime	`ROCm4.3.0;ApacheTVM0.8.0;NVIDIA` extension CUDA10.2
Workload	DISB A-E,uniform RT except Poisson E; closed-loop BE; `ResNet152,VGG19,YOLOv3,BERT-base,GPT2` (128 tokens,GPT2 prefill). RT rates50%,100% or10% solo-throughput according to workload; Apollo/SVL trace. Separate LLM case batch1,100 decode tokens,MLC-LLM.
Tenants	DISB `A/B:1RT+1BE;C:1RT+5BE;D/E:5RT+5BE`
Load and SLO	Table2 `RT100/220/100/20/20req/s;D` uniform,E Poisson;BE closed-loop;
Comparator	RT-only; sequential priority; GPU streams. NVIDIA closed-source strata compare priority-enabled Triton/TensorRT and Orion/PyTorch without dynamic padding.
Finding	GV100 PyTorch DISB-C: Orion p99 normalized latency1.17x vs Reef below1.05x; normalized total throughput0.97x vs1.14x(Fig29). TensorRT comparison trades throughput: mean1.21x Triton vs1.11x Reef(Fig28). All ratios retain model-solo normalization; do not mix engines/AMD strata.
Performance isolation	Per-request normalized RT distributions/p99 and separately normalized total throughput; engine/stratum retained
Security boundary	Not established by scheduler priority
Fault boundary	Kernel killing/restart is a scheduling mechanism, not adversarial fault containment
Overhead	Conference REAL workload records RT end-to-end overhead below 2%; individual evidence rows retain workload-specific values. Journal companion comparisons use separate engines and normalization.

Field	Extracted content
Limitations	Conference outcomes and original QA remain OSDI 2022 anchored; journal observations are explicitly labelled companion results. Deterministic/stateless kernel assumption; closed-source comparison omits dynamic padding. Sufficient memory and bandwidth assumptions constrain transfer. Both versions map to one study identity. All retained E018-E026 values are `conferenceP005/AMD-stratum` outcomes. P023 is linked companion context; no journal-only NVIDIA/LLM results are spliced into these rows. All retained E018-E026 values are `conferenceP005/AMD-stratum` outcomes. P023 is linked companion context; no journal-only NVIDIA/LLM results are spliced into these rows.
Source	https://www.usenix.org/system/files/osdi22-han.pdf
Evidence location	PDF pp11-12 / printed548-549; Table2; Figure11; section7.2

QA	Score / maximum	Rationale and anchor
Q1	2 / 2	Concurrent RT/BE DNN inference with low latency and work conservation is explicit.. PDF p2
Q2	2 / 2	Reset preemption, idempotent re-execution, queues and dynamic kernel padding are described.. PDF pp4-11; sections3-6
Q3	1 / 2	DISB scenarios and baseline methods clear; end-to-end repeated-run/variance protocol not established.. PDF pp11-12; sections7.1-7.2
Q4	2 / 2	MI50,CPU,16GB host `RAM,Ubuntu18.04,ROCm4.3.0,TVM0.8.0` and clients supplied.. PDF p11; Table2
Q5	2 / 2	RT-only,SEQ,GPUStreams and extended wait-based preemption conditions are explicit.. PDF pp11-12; sections7.1-7.3
Q6	2 / 2	Figure11 throughput includes both RT and BE clients; benchmark/load is retained for each ratio.. PDF p12 / printed549; Figure11
Q7	1 / 2	End-to-end RT overhead is stated with offered loads; request percentile for retained headline overheads unresolved.. PDF pp11-12; sections7.1-7.2

QA	Score / maximum	Rationale and anchor
Q8	2 / 2	Single/multiple RT and BE clients under uniform,Poisson and trace workloads are tested.. PDF pp11-12; Table2
Q9	2 / 2	Q9=2 for explicitly defined priority/performance protection and direct latency-interference tests; journal-only NVIDIA results remain separate.. Conference PDF pp2-3,8-11; sections2 and5
Q10	2 / 2	Artifact,platform and idempotence constraints disclosed; conference REEF-N is distinct from later journal NVIDIA implementation.. PDF pp10-11,15-16; P023p1

B.6 S006 / P006 Paella: Low-latency Model Serving with Software-defined GPU Scheduling

[18]

Field	Extracted content
Mechanism	Temporal software scheduling + MPS comparison
Hardware and environment	TeslaT4; XeonSilver4114/64GB; Linux4.15.0; CUDA11.7; modified TVM0.10
Runtime	`Linux4.15.0;CUDA11.7;TVM0.10.0`
Workload	Eight models; lognormal sigma1.5/2; stabilized measurements
Tenants	Mixed inference; lognormal arrivals sigma1.5/2
Load and SLO	Bursty lognormal arrivals with sigma 1.5/2, eight models; evaluates p99 latency and throughput. A common fixed numerical SLO is not established in the current extract.
Comparator	CUDA single/multiple streams; MPS; Triton23.03; ablations
Finding	p99 latency and throughput under bursty model mixes; Fig12 includes MPS.
Performance isolation	Evaluates p99 latency and throughput under mixed inference loads; Fig. 12 includes MPS. This does not establish a universal latency guarantee.
Security boundary	Not assessed for adversarial security in the retained source evidence; performance-sharing results do not establish memory/security isolation.
Fault boundary	Not assessed for cross-tenant fault containment in the retained source evidence; scheduling/performance results do not establish it.
Overhead	Compiler/runtime integration: modified TVM and a custom Triton TVM wrapper. A separate numerical integration/dispatch overhead is not assessed in the retained extract.
Limitations	Custom Triton TVM wrapper; MPS omitted from Fig11; dynamic batching unsupported.
Source	https://vincen.tl/files/ng23paella.pdf
Evidence location	PDF p10 / printed604; section7.1; Figure12 onPDFp11; PDF p12 / printed606; section7.3; Figure15

QA	Score / maximum	Rationale and anchor
Q1	2 / 2	Fine-grained software scheduling for low-latency inference is explicit.. PDF p1
Q2	2 / 2	Instrumented kernel placement/completion notifications,dispatcher and SRPT/deficit policy are described.. PDF pp5-9; sections4-6
Q3	1 / 2	Stabilization before measurement stated; repetitions and uncertainty for headline curve not established.. PDF p10; Methodology
Q4	2 / 2	`T4,CPU,RAM,Linux4.15,CUDA11.7,TVM0.10` and model list supplied.. PDF pp9-10; Table2
Q5	1 / 2	TritonTVM wrapper,omitted Clockwork curves and MPS process-count restriction require qualifications.. PDF p10; Table3; footnote3
Q6	2 / 2	Average throughput versus P99 curves use increasing load until saturation.. PDF pp10-11; Figures11-12
Q7	2 / 2	Up-to3x short-job P99 reduction is author-reported;6.6us is instrumentationP90 overhead, not request latency.. PDF pp10-12; Figures12,15
Q8	2 / 2	Mixed short/long jobs and deficit threshold show latency/fairness trade-offs.. PDF pp10-12; Figures12-13
Q9	N/A	No separately appraised resource/security/fault isolation guarantee; retain reasoned N/A.. PDF pp9-12; scheduling and discussion
Q10	2 / 2	Compiler modifications,versioned setup,CPU dispatch and dynamic-batching limitations described.. PDF pp9-13; sections7-8

B.7 S007 / P007 Orion: Interference-aware, Fine-grained GPU Sharing for ML Applications

[19]

Field	Extracted content
Mechanism	CUDA API interception + temporal
Hardware and environment	V10016GB/CUDA10.2; A10040GB/CUDA11.3; PyTorch1.12; Python3.9
Runtime	`PyTorch1.12;Python3.9;CUDA10.2/11.3`
Workload	Inference-inference and inference-training; uniform/Poisson; three repeats
Tenants	High-priority plus best-effort inference;Poisson
Load and SLO	High-priority plus best-effort inference; uniform/Poisson arrivals and three repeats. Model-specific target values are not assessed in the retained extract.
Comparator	MPS; streams; temporal; REEF-N reimplementation
Finding	Interference-aware scheduling compares inference latency and BE throughput.
Performance isolation	Interference-aware scheduling evaluates high-priority inference latency alongside best-effort throughput against MPS, streams, temporal scheduling and REEF-N.
Security boundary	Not assessed for adversarial security in the retained source evidence; performance-sharing results do not establish memory/security isolation.
Fault boundary	Not assessed for cross-tenant fault containment in the retained source evidence; scheduling/performance results do not establish it.
Overhead	CUDA interception and scheduling integration; clients in the prototype share a process. A standalone numerical scheduling overhead is not assessed in the retained extract.
Limitations	Prototype clients share a process; REEF-N differs from AMD REEF. P99effect is source-described about2x; do not claim graph-extracted precision. P99effect is source-described about2x; do not claim graph-extracted precision.
Source	https://fotstrt.github.io/files/2024-orion.pdf
Evidence location	PDF p12 / printed1086; section6.2.3; Figure12

QA	Score / maximum	Rationale and anchor
Q1	2 / 2	Interference-aware fine-grained local GPU sharing is explicit.. PDF p1
Q2	2 / 2	CUDA interception,per-client queues,resource profiling and throttling are specified.. PDF pp5-8; sections4-5
Q3	2 / 2	Three experiment repetitions and interpretation of variance bars are specified.. PDF pp8-9,12; sections6.1,6.3
Q4	2 / 2	`V10016GB/A10040GB,VM,PyTorch1.12,Python3.9,CUDA10.2/11.3` and models supplied.. PDF pp4,8-9; Tables1,3
Q5	1 / 2	REEF-N and Tick-Tock are reimplemented; Streams threads and MPS processes differ.. PDF p9; Baselines
Q6	2 / 2	7.3x is aggregate throughput maximum versus GPU dedicated only to high-priority inference.. PDF p12; section6.2.3
Q7	2 / 2	Poisson15% and Apollo22% overhead summaries are distinct; Figure12 averages across co-locations.. PDF pp9,12; Figures11-12
Q8	2 / 2	Two-client tests and five-clientA100 experiment test interference at identified scales.. PDF pp9,12; sections6.2-6.3
Q9	2 / 2	Same-trust-domain assumption explicitly limits isolation; direct performance evidence does not imply security.. PDF p13; Security
Q10	2 / 2	Interception/profiling cost,cache interference and cluster co-design limits are explicit.. PDF p13; sections6.5,7

B.8 S008 / P008 GSLICE: Controlled Spatial Sharing of GPUs for a Scalable Inference Platform

[24]

Field	Extracted content
Mechanism	MPS adaptive provisioning
Hardware and environment	One V10016GB; XeonGold6148; kernel4.4.0-150; DPDK19.02; CNTK/PyTorch/TensorRT
Runtime	`Linux4.4.0-150;DPDK19.02;CNTK/PyTorch/TensorRT`
Workload	Two-four inference functions; VGG case SLO50ms; traffic replay
Tenants	Concurrent inference functions;25ms target
Load and SLO	Two to four concurrent inference functions and traffic replay; recorded targets include a 25 ms case and a VGG case with 50 ms SLO. Keep workload-specific targets separate.
Comparator	Standalone functions; MPS provisioning variants
Finding	GUE measures efficiency; shadow instance loading takes about12s in Fig15.
Performance isolation	Adaptive MPS provisioning and measured function latency/efficiency; median-latency results are not a p99 isolation guarantee.
Security boundary	Not assessed for adversarial security in the retained source evidence; performance-sharing results do not establish memory/security isolation.
Fault boundary	Not assessed for cross-tenant fault containment in the retained source evidence; scheduling/performance results do not establish it.
Overhead	Shadow-instance loading takes approximately 12 s in Fig. 15; loading time is not steady-state request latency.
Limitations	GUE is not GPU busy percentage; median latency not p99; academic mirror.
Source	https://www.cse.iitb.ac.in/approximately `puru/courses/spring2024-25/downloads/gslice.pdf`
Evidence location	GSLICE PDF p9; printed p500; Table3; GSLICE PDF p10; printed p501; Table4 and section4.3

QA	Score / maximum	Rationale and anchor
Q1	2 / 2	Inference sharing objective explicit.. PDF p1; printed p492; abstract/introduction
Q2	2 / 2	MPS allocation and adaptive batching described.. PDF pp4-8; section3
Q3	1 / 2	Confidence intervals shown; repeated-run count not specified for retained results.. PDF p9; printed p500; section4.1
Q4	2 / 2	V100 testbed, kernel, frameworks and concurrent models specified.. PDF pp8-9; section4 setup
Q5	2 / 2	TensorRT comparison matches GPU-ready batch execution and states excluded transport costs.. PDF p10; printed p501; section4.3
Q6	2 / 2	GUE defined as an efficacy metric; table values are not busy percentages.. PDF p3; printed p494; PDF p9 Table3
Q7	1 / 2	Median latency and SLO shown; request-tail quantile not established.. PDF p10; section4.4; Figures12a-12b
Q8	2 / 2	Two-to-four-function combinations evaluated.. PDF p9; section4.2; Figure10
Q9	2 / 2	Performance boundary evaluated under allocated GPU shares; no adversarial guarantee inferred.. PDF pp5-7,10-11; sections3.2,4.5
Q10	1 / 2	Implementation and sensitivity described; full reproduction package not established in accessed copy.. PDF pp8-13; sections4-6

B.9 S009 / P009 Serving DNNs like Clockwork: Performance Predictability from the Bottom Up

[20]

Field	Extracted content
Mechanism	Temporal exclusive scheduling
Hardware and environment	V10032GB; DellR740; worker32cores/768GB; two GPUs/host
Runtime	TVM v0.7 compiler and custom managed runtime reported (PDF pp4,7-8); exact OS/driver versions not established in setup.
Workload	Local experiment: one GPU; 15 RN50 copies; 16 clients/model; SLO10-500ms
Tenants	15 ResNet50 instances;16 closed-loop clients/model
Load and SLO	One-GPU local experiment: 15 ResNet50 instances, 16 closed-loop clients per model, SLO range 10-500 ms; Fig. 5 reports SLO-compliant goodput.
Comparator	Clockwork variants; INFaaS on different hardware
Finding	Fig5 measures goodput satisfying SLO, rather than raw request throughput.
Performance isolation	Exclusive temporal scheduling and admission control aim at predictable completion. Goodput counts SLO-compliant requests; no universal cross-workload p99 guarantee is inferred.
Security boundary	Not assessed for adversarial security in the retained source evidence; performance-sharing results do not establish memory/security isolation.
Fault boundary	Not assessed for cross-tenant fault containment in the retained source evidence; scheduling/performance results do not establish it.
Overhead	Not assessed as a separate numerical overhead in the retained source extract; controller/worker costs and baseline hardware must not be conflated.
Limitations	Cross-hardware INFaaS comparison is indirect; controller/worker distinction.
Source	https://www.usenix.org/system/files/osdi20-gujarati.pdf

Field	Extracted content
Evidence location	USENIX PDF pp10-11; printed pp451-452; Figure5 and section6.1

QA	Score / maximum	Rationale and anchor
Q1	2 / 2	Predictable multi-model inference objective and target behavior explicit.. PDF p2; printed p443; section1
Q2	2 / 2	Controller scheduling, worker actions and managed model runtime described.. PDF pp6-10; sections4-5
Q3	1 / 2	Section6.6 reports three scaling repetitions; repeated-run count for retained Figure5 is not specified.. PDF pp10-11,14; sections6.1,6.6
Q4	2 / 2	V100 hardware, TVM v0.7 compiler, model set and client configuration are reported.. PDF pp4,7,10; Table1; sections2,6
Q5	1 / 2	INFaaS uses different EC2 machines; authors restrict that contrast to qualitative interpretation.. PDF p10; printed p451; section6.1
Q6	2 / 2	Goodput counts successful requests within their deadline and excludes late requests.. PDF p11; printed p452; section6.1
Q7	2 / 2	SLO range, closed-loop load, latency CDF and batch scheduling explanation are explicit.. PDF pp10-11; Figure5; section6.1
Q8	2 / 2	Concurrent model demand and a protected minor workload tested.. PDF pp10-12; sections6.1-6.3; Figures5-7
Q9	2 / 2	Performance-isolation boundary is measured; dedicated-worker assumptions and security exclusions are explicit.. PDF pp11-12,14-15; sections6.2,7
Q10	2 / 2	Runtime, scheduling, model details and limitations support informed implementation appraisal; no local reproduction claimed.. PDF pp7-10,14-15; sections5,7

B.10 S010 / P010 Colocating ML Inference and Training with Fast GPU Memory Handover (SIRIUS)

[27]

Field	Extracted content
Mechanism	Temporal + MPS memory handover
Hardware and environment	V10016GB; Ubuntu20.04; CUDA11.6; PyTorch2.1.2; TVM0.14; LLM case A10080GB
Runtime	Python3.10; CUDA11.6; PyTorch2.1.2; TVM0.14;
Workload	One GPU; six DNNs/up to56 instances + SWIN training; 300s; Poisson MAF; SLO4x solo
Tenants	six inference models duplicated round-robin to56 instances; co-located Swin-T training, batch72
Load and SLO	Light/Heavy/Burst/Skewed and compressed Azure Functions traces; 300s runs; SLO=4x standalone inference execution; Tidle5s, watermark1GB
Comparator	Reimplemented TaskSwitch; SP50/SP75; UM+MPS
Finding	Inference p99/SLO outcomes accompany training progress; separate TTFT/TBT LLM case.
Performance isolation	Inference p99/SLO compliance evaluated alongside best-effort training progress; SLO is four times standalone inference execution in the recorded trace tests.
Security boundary	Safe memory reclamation/handover is a design property. Adversarial cross-tenant memory isolation is not established by the retained performance evidence.
Fault boundary	Safe memory handover is assessed as runtime correctness; cross-tenant fault containment is not established by the retained evidence.
Overhead	memory adjustment/allocation and batch discard evaluated in section6.3;
Limitations	TaskSwitch is a reimplementation, not vanilla PipeSwitch.
Source	https://www.usenix.org/system/files/atc25-wang-jiali.pdf
Evidence location	USENIX PDF p11; printed p1666; Figure10 discussion

QA	Score / maximum	Rationale and anchor
Q1	2 / 2	Inference-priority co-location objective and training progress goal explicit.. PDF p2; printed p1657; abstract/introduction
Q2	2 / 2	VMM pool, handover, memory mapping and SM allocation described.. PDF pp5-10; sections3-5
Q3	1 / 2	Trace distributions and 300-second window specified; repeated-run uncertainty not specified for Figure10.. PDF pp10-11; section6; Figure10
Q4	2 / 2	V100/A100 cases, CPU/RAM, software versions and model/batch table supplied.. PDF pp10-11,14; Table3; sections6,6.6
Q5	1 / 2	TaskSwitch is an adapted implementation; equivalence to unmodified PipeSwitch is not established.. PDF p10; comparing targets
Q6	2 / 2	Training samples/second and inference SLO compliance are defined separately.. PDF p11; metrics; Figure10
Q7	2 / 2	P99, arrivals, duration and four-times-solo SLO definition stated.. PDF p11; section6.1
Q8	2 / 2	Inference plus Swin-T training co-location evaluated on a single GPU.. PDF p11; Figure10
Q9	1 / 2	Handover correctness and priority boundary described; direct fault-injection validation not established.. PDF pp6-10; sections4-5
Q10	2 / 2	Artifact appendix reports availability, functionality and reproduction badges with artifact DOI.. PDF p20; artifact appendix; `DOI10.5281/zenodo.15581800`

B.11 S011 / P011 ParvaGPU: Efficient Spatial GPU Sharing for Large-Scale DNN Inference in Cloud Environments

[1]

Field	Extracted content
Mechanism	MIG + MPS
Hardware and environment	A10080GB; Ubuntu20.04; CUDA12.0.1; reported PyTorch1.14.0 needs checking
Runtime	CUDA12.0.1; PyTorch1.14.0 as printed in source (unverified version string);
Workload	Eleven models; six scenario mixes; same-model MPS inside MIG
Tenants	one to three homogeneous MPS processes within a MIG instance; other models can occupy separate segments
Load and SLO	extracted example:1-GPC MIG instance; batch4;1/2/3 MPS processes; no tail quantile identified for local latency
Comparator	MIG-serving; gpulet and system variants
Finding	Fig7 separates internal/external fragmentation; Fig5 counts GPUs.
Performance isolation	MIG/MPS configurations evaluated against workload-specific SLOs and GPU usage; local batch-4 latency example does not identify a tail percentile.
Security boundary	MIG partitioning is an architectural boundary; a dedicated adversarial security assessment is not established in the retained extract.
Fault boundary	Cross-tenant fault containment is not assessed in the retained source extract; GPU counts and SLO results do not establish it.
Overhead	scheduling/profiling discussed;
Limitations	Author manuscript; unusual runtime version; GPU count is not occupancy.
Source	https://arxiv.org/pdf/2409.14447
Evidence location	arXiv v1 PDF p5; sectionIII-B; Figures3-4 on p4

QA	Score / maximum	Rationale and anchor
Q1	2 / 2	SLO-constrained inference resource objective explicit.. arXiv v1 PDF pp1-2
Q2	2 / 2	MIG/MPS profiling, segment configuration and allocation algorithms described.. arXiv v1 PDF pp4-8; sectionIII
Q3	1 / 2	Profiler and six scenarios specified; repeated-run uncertainty not established.. arXiv v1 PDF pp5,8-10; sectionsIII-C,IV
Q4	1 / 2	A100, Ubuntu and CUDA reported; printed PyTorch1.14.0 string is an unresolved environment ambiguity.. arXiv v1 PDF p8; sectionIV-A
Q5	1 / 2	Local process-count contrast interpretable; fleet baseline variants and manuscript scope constrain comparison.. arXiv v1 PDF pp5,8; sectionsIII-B,IV-A
Q6	2 / 2	Local throughput values and metric units given.. arXiv v1 PDF pp4-5; Figure3; sectionIII-B
Q7	1 / 2	Local latency values given without a named tail quantile.. arXiv v1 PDF pp4-5; Figure4; sectionIII-B
Q8	2 / 2	One-to-three MPS processes measured inside a MIG segment.. arXiv v1 PDF pp4-5; Figures3-4
Q9	1 / 2	MIG/MPS boundary described; no dedicated fault or adversarial experiment established in reviewed manuscript.. arXiv v1 PDF pp2-4,8-10
Q10	1 / 2	Algorithms and reconfiguration limitation described; accessed manuscript has no verified final-version equivalence.. arXiv v1 PDF pp5-10; sectionIII-F; source/version record

B.12 S012 / P012 Torpor: GPU-Enabled Serverless Computing for Low-Latency, Resource-Efficient Inference

[28]

Field	Extracted content
Mechanism	API remoting + temporal swapping
Hardware and environment	Worker48vCPU/384GB; four V10032GB
Runtime	PyTorch-based clients replace CUDA libraries; exact OS/driver/CUDA versions not reported in section7 settings
Workload	Eight inference models; warm-up; CV80ms/BERT200ms SLO
Tenants	multiple low-frequency ResNet152 functions share one GPU; sufficient functions to saturate device memory
Load and SLO	10-80 `requests/minute/function;` global defaultP98; CV deadline80ms and Bert-qa200ms
Comparator	Native; remote GPU; PCIe/NVLink swap
Finding	BERT Table4: native42ms, remote43ms, PCIe swap144ms, NVLink swap45ms.
Performance isolation	P98 response-time/SLO results evaluated under low-frequency serverless inference, with CV 80 ms and BERT-QA 200 ms deadlines; remoting and swapping conditions remain separate.
Security boundary	Section4.5 describes shared and isolated GPU runtimes; section7 evaluates runtime sharing, while section8 deployment uses isolation mode. No universal adversarial-security guarantee inferred.
Fault boundary	Section4.5 describes restart, migration and persisted-state recovery. Direct fault-injection validation is not established by the retained performance results.
Overhead	API redirection, swapping and extra CPU activity evaluated; ResNet152 remoting17ms versus Native25ms, Table4
Limitations	p98 tail; busy time is not SM occupancy; node scaling separate.
Source	https://www.usenix.org/system/files/atc25-yu.pdf
Evidence location	USENIX PDF pp10-11; printed pp605-606; Figure8 and section7.2; USENIX PDF pp10-11; printed pp605-606; default P98 metrics and section7.2

QA	Score / maximum	Rationale and anchor
Q1	2 / 2	Low-frequency serverless inference density and latency objectives explicit.. PDF p2; printed p597; section1
Q2	2 / 2	Remoting, swapping, memory manager, scheduler and two runtime modes described.. PDF pp5-9; sections4-6
Q3	1 / 2	Trace, rates and warm-up specified; repeated-run uncertainty not specified for retained Figure8.. PDF pp9-11; section7
Q4	1 / 2	Worker and model configuration supplied; exact OS/driver/framework version matrix not specified in setup.. PDF p9; printed p604; settings
Q5	1 / 2	Native and adapted INFless-KA baselines described; CPU execution paths and adaptations affect equality.. PDF pp10-11; sections7.1-7.3
Q6	2 / 2	Normalized single-GPU throughput and GPU busy-time definition supplied.. PDF pp10-11; metrics; Figure8
Q7	2 / 2	Default P98, per-function rates and deadlines stated.. PDF pp10-11; metrics; section7.2
Q8	2 / 2	Many low-frequency functions share one GPU; node-level results identified separately.. PDF pp10-11; Figure8; section7.2
Q9	1 / 2	Runtime-sharing and runtime-isolation modes plus recovery design stated; dedicated fault-injection evaluation not established.. PDF p7; section4.5; PDF p9 evaluation mode
Q10	1 / 2	Implementation and mode trade-offs explained; complete public replication recipe not established in paper.. PDF pp7-9,13; sections4.5,6,8

B.13 S013 / P013 StreamBox: A Lightweight GPU SandBox for Serverless Inference Workflow

[29]

Field	Extracted content
Mechanism	API/runtime sandbox virtualization
Hardware and environment	Four V10016GB; 28cores/128GB; CUDA10.1; PyTorch1.3; TensorFlow2.12
Runtime	`PyTorch1.3.0,TensorFlow2.12,CUDA10.1` as reported; driver/OS version not reported in section9 setup
Workload	Three concurrent inference workflows/GPU
Tenants	concurrent functions within workflow share runtime; separate3/6-user Traffic test on one GPU
Load and SLO	Azure Functions7-day invocation trace; `sporadic/periodic/bursty;` Traffic SLO200ms; up to3 workflows/runtime contexts per GPU in evaluated setup
Comparator	MPS; INFless; Astraea; Stream-only
Finding	Table2 evaluates runtime sandbox workflows on shared GPUs.
Performance isolation	Concurrent workflow/runtime configurations evaluated with traffic SLO 200 ms and separate 3/6-user tests. This is workload-specific performance evidence.
Security boundary	Runtime sandbox memory separation is discussed; the retained extract does not establish adversarial security.
Fault boundary	Fault-containment behavior is not assessed in the retained source extract; sandbox terminology alone is not a fault test.
Overhead	API `forwarding1-3microseconds;3-5microseconds` at48functions; one-time profiling36s and transformation4s, p70
Limitations	Sandbox memory separation is not proof of adversarial security.
Source	https://www.usenix.org/system/files/atc24-wu-hao.pdf
Evidence location	USENIX PDF p11; printed p68; section9.1; Figure13; USENIX PDF p13; printed p70; section9.3; Figure18

QA	Score / maximum	Rationale and anchor
Q1	2 / 2	Workflow runtime-redundancy and spatial-sharing objective explicit.. PDF p2; printed p59; section1
Q2	2 / 2	API proxy, PTB transformation, stream sharing and memory pool described.. PDF pp5-10; sections3-8
Q3	1 / 2	Trace types, ablations and workloads stated; repeated-run uncertainty not specified for Figure13.. PDF pp10-13; section9
Q4	2 / 2	V100 hardware, CPU/RAM, framework versions, CUDA version and workflow descriptions reported.. PDF p10; printed p67; setup; Table2
Q5	1 / 2	Warm-up and common optimizations disclosed; adapted baselines and batch/concurrency choices constrain causal attribution.. PDF pp10-11; comparing targets; Figure13
Q6	2 / 2	Throughput, memory metrics and comparator-specific factors explicitly reported.. PDF p11; printed p68; Figures12-13
Q7	1 / 2	Traffic SLO and CDF supplied; exact scalar request-tail quantile not supplied for retained throughput result.. PDF p11; Figure14
Q8	2 / 2	Separate three- and six-user single-GPU tests supplied.. PDF p13; section9.4; Figure19
Q9	2 / 2	Mutual trust within workflow is explicit; separate proxies for users are distinguished from untested strong security.. PDF pp5-6,13; sections3-4,9.4,10
Q10	2 / 2	Implementation, preparation costs and multi-user limitations sufficiently detailed for informed appraisal.. PDF pp6-10,13; sections4-9

B.14 S014 / P014 Efficient Performance-Aware GPU Sharing with Compatibility and Isolation through Kernel Space Interception (KRYPTON)

[30]

Field	Extracted content
Mechanism	Kernel API virtualization + hybrid
Hardware and environment	A10040GB/RTX4090; Ubuntu20.04; kernel5.15.0-91; driver550.90.07; CUDA12.1; PyTorch2.2.1
Runtime	`NVIDIA550.90.07;Docker24.0.7;CUDA12.1;Vulkan1.3;PyTorch2.2.1`
Workload	`Inference/training/rendering` in separate containers; solo-throughput targets
Tenants	inference with co-running applications; separate training/rendering results not substituted
Load and SLO	QoS: initial Poisson rate is half solo maximum; shared setting halves the input again and assigns50%time; tokens5/10/20/40/80ms; target original soloP99.
Comparator	Temporal; MIG; adapted gpulet
Finding	Inference-specific sharing comparisons; section2.2 tests fault propagation.
Performance isolation	Inference QoS uses solo-p99 targets under Poisson load at half maximum throughput; temporal allocation/token settings and adapted baselines are retained.
Security boundary	Container/kernel interception and partition boundaries are design features. A dedicated adversarial security assessment is not established by the retained extract.
Fault boundary	Section 2.2 Fig. 3 examines fault propagation in sharing baselines. This observation is not treated as proof of universal KRYPTON fault containment.
Overhead	token-length trade-off and4.8MBhost-memory module overhead
Limitations	Baseline adaptations and workload mix constrain attribution.
Source	https://www.usenix.org/system/files/atc25-zhang-shulai.pdf
Evidence location	USENIX PDF pp12-13; printed pp1013-1014; section5.5; Figure17; USENIX PDF p13; printed p1014; section5.7

QA	Score / maximum	Rationale and anchor
Q1	2 / 2	Compatibility, isolation and performance-control objectives explicit.. PDF p2; printed p1003; section1
Q2	2 / 2	Kernel interception, token control and MIG orchestration explained.. PDF pp5-10; sections3-4
Q3	1 / 2	Setup and workloads detailed; repeated-run uncertainty not specified for retained QoS result.. PDF pp10-13; section5
Q4	2 / 2	OS, kernel, driver, runtimes, GPUs and workloads explicitly tabulated.. PDF p10; printed p1011; Tables1-2
Q5	1 / 2	Isolation-capable baselines and MIG adaptations described; policy and allocation differences limit equivalence.. PDF pp10-11; section5.1
Q6	2 / 2	Throughput targets are solo-normalized and QoS violation is a request fraction.. PDF pp10,12-13; sections5.1,5.4-5.6
Q7	2 / 2	Solo-P99 threshold, Poisson load, subsequent rate halving and token sizes stated.. PDF pp12-13; section5.5; Figure17
Q8	2 / 2	Co-running inference relative-error and QoS experiments supplied.. PDF p13; sections5.5-5.6
Q9	1 / 2	Baseline fault propagation is tested; design/security argument does not directly establish universal KRYPTON containment.. PDF p4 Figure3; PDF p14 section7
Q10	1 / 2	Implementation and MIG limitation detailed; complete end-to-end replication package not established.. PDF pp10,14; sections4,7

B.15 S015 / P015 XSched: Preemptive Scheduling for Diverse XPUs

[2]

Field	Extracted content
Mechanism	Temporal API interposition
Hardware and environment	One GV100 inference case; broader XPU experiments separate
Runtime	CUDA/Triton and Paella artifact; exact version numbers not reported in extracted case3
Workload	Two BERT-large clients; HP10req/s; saturated BE
Tenants	one high-priority client `at10requests/s;low-priority` client continuous on sameGV100
Load and SLO	Triton client rates as above;Paella workload lognormal sigma2.0;K-EDF K16; exact batch sizes not reported in extracted case3
Comparator	Triton; Triton+priority; Triton+XSched
Finding	Fig15: relative p99 1.53/1.51/1.07 respectively versus solo.
Performance isolation	Two BERT-large clients on one GV100: Fig. 15 gives relative p99 1.53/1.51/1.07 for `Triton/Triton+priority/Triton+XSched` versus solo.
Security boundary	Scheduling API interposition does not establish adversarial memory isolation; security is not assessed in this extracted inference case.
Fault boundary	Cross-tenant fault containment is not assessed in the extracted inference case; device/preemption limitations remain separate.
Overhead	10lines Triton integration and15lines Paella-baseline integration;preemption overhead measured separately
Limitations	Data assumed to fit memory; CUDA graphs and undocumented driver limits. XSched GPU Triton case3 is Section8, PDFp14/printed683; NPU/ASIC/FPGA outcomes excluded. XSched GPU Triton case3 is Section8, PDFp14/printed683; NPU/ASIC/FPGA outcomes excluded.
Source	https://www.usenix.org/system/files/osdi25-shen-weihang.pdf
Evidence location	PDF p14; printed p683; section8 case3; Figure15a

QA	Score / maximum	Rationale and anchor
Q1	2 / 2	Portable preemptive scheduling objective and GPU-serving case are explicit.. PDF pp2-3; section1
Q2	2 / 2	XQueue, XShim and three preemption levels specify the mechanism.. PDF pp4-10; sections3-6
Q3	1 / 2	Setup and policies are described; repeated-run uncertainty for Figure15 is not specified.. PDF pp11-14; sections7-8
Q4	1 / 2	GV100, BERT-large and client rates are explicit; exact Triton/CUDA versions and batch sizes for case3 remain unresolved.. PDF p14; section8 case3
Q5	2 / 2	Same-GPU standalone, native and priority Triton cases use the stated client conditions.. PDF p14; Figure15a
Q6	2 / 2	Throughput in requests/s and normalized P99 comparisons are distinguished.. PDF p14; Figure15
Q7	2 / 2	P99, latency CDF, high-priority 10 requests/s and continuous background load are explicit.. PDF p14; section8 case3; Figure15a
Q8	2 / 2	Two BERT-large clients share one GV100 in the retained experiment.. PDF p14; section8 case3
Q9	2 / 2	Direct foreground performance-control evidence is delimited; malicious bypass is explicitly a limitation, not a security guarantee.. PDF pp14-15; Figure15a; section9 Untrusted tenants
Q10	2 / 2	Artifact instructions, hardware-dependent preemption limits and sufficient-memory assumption are documented.. PDF pp15,23; section9; Artifact Appendix

B.16 S016 / P018 InferFair: Towards QoS-aware scheduling for performance isolation guarantee in heterogeneous model serving systems

[34]

Field	Extracted content
Mechanism	Temporal SLO-aware scheduling + feedback goodput balancing; MPS comparator
Hardware and environment	Dell R740 host; NVIDIA V10032GB; precise CPU,OS,driver/CUDA versions not established.
Runtime	Clockwork-based engine and InferFair estimator/feedback scheduling; no local replication claimed.
Workload	Table3 local example: three models R2,R3,R4; evaluated 2/3-model local experiments are separate from 4-GPU61-modelMAF trace.
Tenants	3models in retained Table3 example
Load and SLO	600requests/s total; per-model ratio3:1:2; Poisson arrivals; SLO100ms; scheduler period50ms.
Comparator	Clockwork; MPS+Clockwork; EDF, with no-batching comparators explicitly justified.
Finding	524.29SLO-compliant requests/s versus Clockwork457.24; max/min slowdown unfairness1.05 versus1.50 in Table3row3:1:2.
Performance isolation	Performance fairness measured as maximum/minimum model slowdown (Eq2); value near1 means more equal slowdown.
Security boundary	Security not established by fairness results
Fault boundary	Fault isolation not directly tested
Overhead	Online feedback/scheduling and batching costs are qualitative here; a standalone numerical scheduling overhead is not assessed in the retained extract.
Limitations	Repetition/uncertainty and exact software versions unreported; fairness is not throughput maximization or a hostile-tenant guarantee.
Source	https://www.sciencedirect.com/science/article/pii/S0167739X23003187
Evidence location	PDFp9 / `publishedp18;Table3;unfairness` definitionPDFp2 / publishedp11;Eq2

Field	Extracted content

QA	Score / maximum	Rationale and anchor
Q1	2 / 2	Performance fairness for heterogeneous inference models and goodput objectives are explicit.. PDF pp1-2; sections1.1-1.4
Q2	2 / 2	Online solo-run goodput estimator, model-state arrays, temporal scheduling and feedback controller are specified.. PDF pp5-8; section4; Algorithms1-3
Q3	1 / 2	Poisson load, sensitivity and trace duration are given; experiment repetitions and uncertainty remain unspecified.. PDF pp8-9; sections5.1-5.4
Q4	1 / 2	`DellR740,V10032GB,models,load` andSLO are supplied; precise `CPU/OS/driver/CUDA/runtime` versions remain unreported.. PDF pp2,8-9; Table1; section5
Q5	2 / 2	Clockwork,MPS+Clockwork and EDF conditions are explained; EDF intentionally lacks batching and is interpreted as a policy contrast.. PDF p8; section5.1
Q6	2 / 2	Goodput is successful requests per second withinSLO; unfairness is max/min solo-to-shared goodput slowdown.. PDF pp2,8-9; Eq1-2; Table3
Q7	2 / 2	`SLO100ms,interval50ms,Poisson` arrivals and saturation beyond1% deadline failures are specified; no numericalP99 latency is inferred.. PDF p9; sections5.2-5.3
Q8	2 / 2	Two- and three-model local co-running experiments are distinguishable from61-model,fourGPU trace replay.. PDF p9; sections5.3-5.4
Q9	2 / 2	Performance-isolation boundary is explicitly defined by unequal goodput slowdown and directly tested; no memory/security/fault guarantee is claimed.. PDF p2; Eq1-2; PDF p9; Table3
Q10	1 / 2	Algorithms and memory-management reliance onClockwork are explicit; versioned reproduction setup/artifact detail remains incomplete.. PDF pp5-8,10; section7

B.17 S017 / P019 ArkGPU: enabling applications' high-goodput co-location execution on multitasking GPUs

[31]

Field	Extracted content
Mechanism	CUDA API interception + NVML monitoring and admission throttling
Hardware and environment	Machine with four V100s; Xeon Platinum8163; CentOS7.9.2009; driver460.106.00; CUDA11.1; cuDNN7.6.5.
Runtime	CentOS 7.9.2009; NVIDIA driver 460.106.00; CUDA 11.1; cuDNN 7.6.5 (recorded setup).
Workload	Five MLPerf inference LC models (ResNet50, SSD-MobileNet, SSD-ResNet34, DLRM, BERT), each paired locally with six BE training models (30 pairs). Single-stream query size1; p99 QoS threshold twice solo. Separate four-GPU app-mixes are not local occupancy evidence.
Tenants	One latency-critical inference model paired with one best-effort training model locally; five inference by six training models give 30 tested pairs.
Load and SLO	MLPerf single-stream, query size 1; inference QoS threshold is twice solo p99. Separate multi-GPU app mixes are not the local test.
Comparator	MPS (Non-ArkGPU); k8s-native for QPS/utilization; GaiaGPU and adjustment-unit ablation.
Finding	30tested co-location pairs meet twice-soloP99 deadline; approximately5%QPS regression in low-contention cases is retained.
Performance isolation	p99 tested against twice-solo QoS; average latency, QPS and goodput separately
Security boundary	Not established by API interception
Fault boundary	Background discussion of MPS faults is not an ArkGPU fault test
Overhead	Polling/API sleeps can add delay; BE checkpoint/restart incurs costs. Low-contention pairs can lose about 5% QPS; this is a measured effect, not a universal implementation-overhead figure.

Field	Extracted content
Limitations	Mixed inference/training qualifies via explicit inference p99/QPS and workload transfer rationale; it does not establish inference-only performance. Polling/API sleeps can add delay; BE checkpoint/restart affects costs. Huge goodput ratios can have near-zero baseline yield. Repetitions/variance and chart coordinates unresolved.
Source	https://doi.org/10.1007/s42514-023-00154-y
Evidence location	PDF p11 / printed p314; section5.2; Figure14 on PDFp12; PDF p12 / printed p315; section5.2; Figure16 on PDFp13

QA	Score / maximum	Rationale and anchor
Q1	2 / 2	Objective is QoS-preserving inference/training co-location.. PDF pp1-2; section1
Q2	2 / 2	NVML monitoring, CUDA API interception, predictor and adjustment are specified.. PDF pp6-11; sections4.1-4.3
Q3	1 / 2	Model pairs and measurements are described; repetitions and uncertainty remain unspecified.. PDF pp11-16; section5
Q4	2 / 2	V100, host, CentOS, CUDA and cuDNN plus five inference and six training models are given.. PDF pp4,11; Tables1-3
Q5	1 / 2	MPS/k8s comparators and workload pairs are relevant; exact MPS allocation quotas remain unresolved.. PDF pp11-12; section5.2; Figures14-17
Q6	2 / 2	QPS, query yield and goodput are distinguished with explicit definitions.. PDF pp12-14; sections5.2-5.3; equations2-3
Q7	2 / 2	Single-stream query size1, P99 and twice-solo QoS target are explicit.. PDF pp4,11-12; sections3.4,5.1-5.2
Q8	2 / 2	Thirty inference/training pairs and adjustment ablation directly test interference.. PDF pp11-16; Figures14-21
Q9	1 / 2	Performance restriction is tested, but stronger cross-context/error claims lack a direct fault/security experiment.. PDF pp2-3,10-11,15-16; sections4.3,5.4
Q10	1 / 2	Prediction/limiter overhead and limitations are discussed; data are on request and artifact/software details remain incomplete.. PDF pp15-16; sections5.5-5.6; data availability

B.18 S018 / P020 Dyna-P: placement-aware dynamic partitioning for lightweight applications with modern GPUs

[25]

Field	Extracted content
Mechanism	MIG dynamic `placement/reconfiguration` with reconfiguration-last policy
Hardware and environment	Two A30s,24GB each; CUDA12.2; driver535.171.04; DCGM3.1.3. Local sections4.2/4.3.1 use one A30. StaCon `profiles2g.12gb+1g.6gb+1g.6gb.`
Runtime	CUDA 12.2; NVIDIA driver 535.171.04; DCGM 3.1.3 (recorded setup).
Workload	50 SLM inference jobs; batch1-64; DistilGPT, Codegen, Flan-T5-Large, Codellama and Phi-1 as labelled in Table2. Local batch sensitivity:10 inference jobs, batches1-8 vs32-64 on one A30. Later160-A100 cluster study is simulation.
Tenants	Multiple SLM inference jobs occupy MIG partitions; local batch experiment has 10 jobs on one A30. Separate 50-job and simulated cluster results are distinguished.
Load and SLO	Inference batch sizes 1-64; local sensitivity compares batches 1-8 versus 32-64. JCT/makespan/STP are job metrics, not online request-tail SLO evidence.
Comparator	Full-GPU sequential baseline; fixed StaCon MIG; pre-profiled MISO adapted to A30; ORACLE with future job/size/time knowledge.
Finding	15%prediction error causes8.09%longer makespan and2additional reconfigurations; Section4.2 local experiment, distinct from Section4.3.2simulation.
Performance isolation	Job completion/makespan/STP and prediction sensitivity; no extracted online p99 guarantee
Security boundary	Architectural isolation discussed, no adversarial test
Fault boundary	OOM observation is not cross-tenant fault containment validation
Overhead	Reconfiguration may stop/restart active jobs; reconfiguration-last placement reduces interruption. A reliable numerical overhead fraction is not assessed from the retained extract.

Field	Extracted content
Limitations	Retain one-GPU inference strata separately from two-GPU aggregate and cluster simulation. JCT/makespan/STP cannot replace request-tail latency or goodput. Reconfiguration can stop/restart active jobs; neither architectural MIG claims nor OOM avoidance prove hostile isolation. Model labels/parameter counts and corrupted overhead fraction require caution.
Source	https://doi.org/10.1007/s10586-025-05284-2
Evidence location	PDF pp13-14; section4.2; Figure12 on PDFp16

QA	Score / maximum	Rationale and anchor
Q1	2 / 2	Fragmentation, placement and dynamic partitioning objective is explicit.. PDF pp1-2; section1
Q2	2 / 2	Partition prediction, BnB scheduling, constraints and reconfiguration-cost model are described.. PDF pp5-11; section3; Algorithms1-3
Q3	1 / 2	Workload counts, comparators and metrics are given; repeated-run uncertainty is not established.. PDF pp11-16; section4
Q4	2 / 2	A30 system, driver, CUDA, DCGM, model variants and batch sizes are specified.. PDF pp11-13; Table2
Q5	1 / 2	MISO is adapted and ORACLE has prior knowledge; comparisons require those qualifications.. PDF p11; section4
Q6	2 / 2	STP is a normalized weighted partition-throughput sum; makespan and JCT are separate.. PDF p11; section4
Q7	1 / 2	Job completion, deadlines and queues are described; online request-tail measurements are not supplied.. PDF pp11-14; sections4.1-4.3
Q8	2 / 2	Single-A30 job/batch concurrency tests are distinguishable from the large-cluster simulation.. PDF pp13-16; sections4.2,4.3.1; Figures11-13
Q9	1 / 2	MIG placement boundaries are defined but no direct `adversarial/fault-containment` test is supplied.. PDF pp8-9,13-14; architectural constraints and local evaluation
Q10	1 / 2	Implementation and `prediction/reconfiguration` limits are disclosed; complete artifact/version details remain unresolved.. PDF pp11,18; sections3.2.5,6.4

QA	Score / maximum	Rationale and anchor

B.19 S019 / P021 KRISP: Enabling Kernel-wise RIght-sizing for Spatial Partitioned GPU Inference Servers

[26]

Field	Extracted content
Mechanism	Spatial partition/API control (AMD)
Hardware and environment	AMD MI50 CU-mask emulation for kernel right-sizing; not a deployable NVIDIA MPS measurement.
Runtime	ROCm 4.5; Ubuntu 18.04; kernel 5.4 (recorded setup).
Workload	One/two/four workers and heterogeneous pairs; batch32; aggregate8/16; eight models
Tenants	One, two and four workers; heterogeneous model pairs on one AMD MI50.
Load and SLO	Recorded tests include batch 32 and aggregate batches 8/16; throughput and p95 inference latency are evaluated. A request-rate/SLO target is not established in this extract.
Comparator	AMD unrestricted concurrency labelled MPSDefault
Finding	Kernel-wise right-sizing evaluates throughput and p95 inference latency.
Performance isolation	Kernel-wise CU right-sizing evaluated through throughput and p95 inference latency. AMD unrestricted concurrency is not NVIDIA MPS.
Security boundary	Not assessed for adversarial security in the retained source evidence; performance-sharing results do not establish memory/security isolation.
Fault boundary	Not assessed for cross-tenant fault containment in the retained source evidence; scheduling/performance results do not establish it.
Overhead	The feature is emulated and emulation overhead is subtracted in reported comparisons. Do not interpret adjusted throughput as measured deployable runtime cost.
Limitations	Feature emulated; overhead subtracted; baseline is not NVIDIA MPS.
Source	https://danielwong.org/files/KRISP-HPCA2023.pdf
Evidence location	Author PDF pp9-11; sectionVI; TableIII and Figure13b

QA	Score / maximum	Rationale and anchor
Q1	2 / 2	Kernel-wise spatial right-sizing for concurrent inference is explicit.. PDF p1; abstract; sectionI
Q2	2 / 2	Kernel-scoped partition proposal and emulation through CU masks are described.. PDF pp5-9; sectionsIV-V
Q3	1 / 2	Emulation correction is transparent; repeated-run uncertainty for headline results is not established.. PDF pp9-11; sectionsV-B,VI
Q4	2 / 2	MI50, host, Ubuntu/kernel, ROCm and model/worker/batch settings are supplied.. PDF pp9-10; sectionVI-A; TableIII
Q5	1 / 2	Model-wise policy permits CU overlap beyond prior isolation constraints; these cases are flagged and limit equivalence.. PDF pp9-10; sectionVI-A
Q6	2 / 2	RPS normalization and energy-per-inference measurement are defined; adjusted results estimate native support.. PDF pp9-11; sectionsV-B,VI-B
Q7	2 / 2	P95 latency and SLO evaluation are explicit under maximum-load inference.. PDF pp10-11; TablesIII-IV; Figure13b
Q8	2 / 2	One, two and four workers plus mixed-model pairs exercise contention.. PDF pp10-12; Figures13-15
Q9	2 / 2	KRISP-I prevents CU overlap and is compared with oversubscribed KRISP-O; this tests a `compute-resource/performance` boundary only.. PDF pp10-12; KRISP-I/O definitions; Figures13,16
Q10	2 / 2	Emulation calibration, adjusted latency formula and proposed hardware changes permit informed appraisal of implementation limits.. PDF pp7-9; sectionsIV-D,V-B

B.20 S020 / P022 Tally: Non-Intrusive Performance Isolation for Concurrent Deep Learning Workloads

[32]

Field	Extracted content
Mechanism	CUDA API interception + priority
Hardware and environment	A10040GB; Ubuntu20.04; CUDA12.2; PyTorch2.2; ONNX1.17
Runtime	ONNXRuntimeGPU1.17.0; inspected arXivv3 copy
Workload	Same-GPU pairs; MAF inference traces; high-priority p99
Tenants	High-priority inference and best-effort training workloads share one A100; model/workload combinations remain experiment-specific.
Load and SLO	MAF-derived inference arrival traces; high-priority p99 compared with standalone execution. The retained throughput contrast uses a 10% idle-time condition.
Comparator	Time slicing; MPS; priority MPS; TGS
Finding	At10%idle-time setting, normalized total system throughput1.19 versus TGS1.65; this is not background-only throughput.
Performance isolation	Thread-block-level scheduling evaluates high-priority inference p99 alongside best-effort throughput; latency protection is not security or fault isolation.
Security boundary	Not assessed for adversarial security in the retained source evidence; performance-sharing results do not establish memory/security isolation.
Fault boundary	Not assessed for cross-tenant fault containment in the retained source evidence; scheduling/performance results do not establish it.
Overhead	At 10% idle time normalized throughput is 1.19 for Tally versus 1.65 for TGS; this is a throughput trade-off, not a standalone integration-overhead measurement.
Limitations	Latency protection sacrifices some BE throughput; author manuscript.
Source	https://arxiv.org/pdf/2410.07381v3
Evidence location	arXiv v3 PDF p10; section5.3; Figure6a on p11

QA	Score / maximum	Rationale and anchor
Q1	2 / 2	Performance isolation for high-priority inference with best-effort co-tenants is explicit.. arXiv v3 PDF pp1-2
Q2	2 / 2	Slicing, preemption, API mediation and profile-guided priority scheduling are described.. arXiv v3 PDF pp5-8; section4
Q3	1 / 2	Profiler averages repeated measurements, but end-to-end repetitions and uncertainty remain unspecified.. arXiv v3 PDF pp8-11; sections4.2,5.1-5.3
Q4	2 / 2	A100 SXM, host, runtime versions and six training/six inference models are explicit.. arXiv v3 PDF p9; Table2; section5.1
Q5	2 / 2	Common workloads, adapted trace, load, hardware and normalized metrics are documented for the four comparison mechanisms.. arXiv v3 PDF pp9-11; sections5.1-5.3
Q6	2 / 2	System throughput sums normalized throughput of all concurrent workloads; it is not background throughput alone.. arXiv v3 PDF pp9-10; sections5.1,5.3
Q7	2 / 2	P99 latency, 50% end-to-end load and 10%-90% idle-time sensitivity are specified.. arXiv v3 PDF pp9-11; Figures5-6
Q8	2 / 2	Inference/training pairs and scaling to multiple inference workloads are evaluated on a shared GPU.. arXiv v3 PDF pp9-11; sections5.2-5.4
Q9	2 / 2	P99 and workload-pair tests directly assess performance isolation; no adversarial memory or cross-tenant fault guarantee is inferred.. arXiv v3 PDF pp9-13; sections5-6
Q10	2 / 2	Artifact setup, `virtualization/kernel-transformation` overhead and CUDA cooperative-kernel limitations are documented.. arXiv v3 PDF pp12-14; sections5.7,6; AppendixA

C Original 52-outcome register

These inherited observations remain unchanged. The 18new observations are in Appendix H.

Reported observations remain attached to their baseline, hardware, workload and source. Values are preserved as supplied, including ranges, approximate values and maxima.

E001 / P001 Throughput ratio

Field	Evidence
Value	2.7
Units	x
Baseline	No MPS
Hardware	NVIDIA A100
Workload	VGG16+GoogLeNet
Interpretation	Batch/rate/quotas unresolved; not tail-latency evidence
Evidence location	PDF pp16-17 / printed458-459; Figure14; section6.7
Source	https://jcst.ict.ac.cn/en/article/pdf/preview/10.1007/s11390-024-3773-5.pdf

E002 / P001 Throughput ratio

Field	Evidence
Value	1.5
Units	x
Baseline	Default MPS
Hardware	NVIDIA A100
Workload	VGG16+GoogLeNet
Interpretation	Same comparison family; not a universal average
Evidence location	PDF pp16-17 / printed458-459; Figure14; section6.7
Source	https://jcst.ict.ac.cn/en/article/pdf/preview/10.1007/s11390-024-3773-5.pdf

E003 / P002 Prediction error range

Field	Evidence
Value	1.53-5.02
Units	percent
Baseline	Measured local latency
Hardware	NVIDIA V100
Workload	`AlexNet/ResNet50/VGG19/SSD;` batch3
Interpretation	Prediction accuracy, not utilization or latency speedup
Evidence location	Final PDF p11 / printed822; Figure13
Source	https://prefer-nsf.org/pdf/IEEE%20TPDS%20_%20igniter%202023.pdf

E004 / P002 Initial p99 latency

Field	Evidence
Value	15.6
Units	ms
Baseline	W1 latency SLO10ms
Hardware	NVIDIA V100
Workload	W1 with W5 and W11
Interpretation	Transient violation precedes recovery; no unconditional guarantee
Evidence location	Final PDF p12 / printed823; Figure17; section5.3
Source	https://prefer-nsf.org/pdf/IEEE%20TPDS%20_%20igniter%202023.pdf

E005 / P004 Additional end-to-end latency

Field	Evidence
Value	6.01
Units	ms
Baseline	Ready-model latency subtracted; MPS overhead307.02ms
Hardware	NVIDIA V10016GB

Field	Evidence
Workload	ResNet152 inference batch8 + training batch32
Interpretation	Mean100 runs; overhead, not total latency or p99; MPS uses unified-memory swapping
Evidence location	PDF pp11-12 / printed508-509; Table3; section6.1
Source	https://www.usenix.org/system/files/osdi20-bai.pdf

E006 / P003 Distribution threshold:90% of measured pairs

Field	Evidence
Value	17.66
Units	percent overhead below threshold
Baseline	Solo-running latency
Hardware	RTX2080Ti11GB
Workload	Paired model inferences
Interpretation	Across-pair CDF, not request-p90/p99; no hostile-memory/fault test
Evidence location	PDF p6 / printed203; Figure7; section3.4
Source	https://www.usenix.org/system/files/atc22-choi-seungbeom.pdf

E007 / P004 Additional end-to-end latency

Field	Evidence
Value	307.02
Units	ms
Baseline	Ready-model latency subtracted
Hardware	NVIDIA V10016GB
Workload	ResNet152 inference batch8 + training batch32
Interpretation	Training task occupies GPU memory and continues. Baseline condition matters; not a universal MPS comparison.
Evidence location	PDF pp11-12 / printed508-509; Table3; section6.1

Field	Evidence
Source	https://www.usenix.org/system/files/osdi20-bai.pdf

E008 / P004 Additional end-to-end latency

Field	Evidence
Value	5.40
Units	ms
Baseline	Ready-model latency subtracted
Hardware	NVIDIA V10016GB
Workload	Inception_v3 inference batch8 + training batch32
Interpretation	Mean of100 runs; overhead not total latency or tail statistic; mixed training/inference workload.
Evidence location	PDF pp11-12 / printed508-509; Table3; section6.1
Source	https://www.usenix.org/system/files/osdi20-bai.pdf

E009 / P004 Additional end-to-end latency

Field	Evidence
Value	232.25
Units	ms
Baseline	Ready-model latency subtracted
Hardware	NVIDIA V10016GB
Workload	Inception_v3 inference batch8 + training batch32
Interpretation	Training task occupies GPU memory and continues. Baseline condition matters; not a universal MPS comparison.
Evidence location	PDF pp11-12 / printed508-509; Table3; section6.1
Source	https://www.usenix.org/system/files/osdi20-bai.pdf

E010 / P004 Additional end-to-end latency

Field	Evidence
Value	10.27
Units	ms
Baseline	Ready-model latency subtracted
Hardware	NVIDIA V10016GB
Workload	Bert_base inference batch8 + training batch32
Interpretation	Mean of100 runs; overhead not total latency or tail statistic; mixed training/inference workload.
Evidence location	PDF pp11-12 / printed508-509; Table3; section6.1
Source	https://www.usenix.org/system/files/osdi20-bai.pdf

E011 / P004 Additional end-to-end latency

Field	Evidence
Value	204.52
Units	ms
Baseline	Ready-model latency subtracted
Hardware	NVIDIA V10016GB
Workload	Bert_base inference batch8 + training batch32
Interpretation	Training task occupies GPU memory and continues. Baseline condition matters; not a universal MPS comparison.
Evidence location	PDF pp11-12 / printed508-509; Table3; section6.1
Source	https://www.usenix.org/system/files/osdi20-bai.pdf

E012 / P004 Additional end-to-end latency

Field	Evidence
Value	5.57
Units	ms
Baseline	Ready-model latency subtracted
Hardware	NVIDIA T416GB

Field	Evidence
Workload	ResNet152 inference batch8 + training batch32
Interpretation	Mean of100 runs; overhead not total latency or tail statistic; mixed training/inference workload.
Evidence location	PDF pp11-12 / printed508-509; Table3; section6.1
Source	https://www.usenix.org/system/files/osdi20-bai.pdf

E013 / P004 Additional end-to-end latency

Field	Evidence
Value	259.20
Units	ms
Baseline	Ready-model latency subtracted
Hardware	NVIDIA T416GB
Workload	ResNet152 inference batch8 + training batch32
Interpretation	Training task occupies GPU memory and continues. Baseline condition matters; not a universal MPS comparison.
Evidence location	PDF pp11-12 / printed508-509; Table3; section6.1
Source	https://www.usenix.org/system/files/osdi20-bai.pdf

E014 / P004 Additional end-to-end latency

Field	Evidence
Value	7.66
Units	ms
Baseline	Ready-model latency subtracted
Hardware	NVIDIA T416GB
Workload	Inception_v3 inference batch8 + training batch32
Interpretation	Mean of100 runs; overhead not total latency or tail statistic; mixed training/inference workload.
Evidence location	PDF pp11-12 / printed508-509; Table3; section6.1

Field	Evidence
Source	https://www.usenix.org/system/files/osdi20-bai.pdf

E015 / P004 Additional end-to-end latency

Field	Evidence
Value	193.05
Units	ms
Baseline	Ready-model latency subtracted
Hardware	NVIDIA T416GB
Workload	Inception_v3 inference batch8 + training batch32
Interpretation	Training task occupies GPU memory and continues. Baseline condition matters; not a universal MPS comparison.
Evidence location	PDF pp11-12 / printed508-509; Table3; section6.1
Source	https://www.usenix.org/system/files/osdi20-bai.pdf

E016 / P004 Additional end-to-end latency

Field	Evidence
Value	34.56
Units	ms
Baseline	Ready-model latency subtracted
Hardware	NVIDIA T416GB
Workload	Bert_base inference batch8 + training batch32
Interpretation	Mean of100 runs; overhead not total latency or tail statistic; mixed training/inference workload.
Evidence location	PDF pp11-12 / printed508-509; Table3; section6.1
Source	https://www.usenix.org/system/files/osdi20-bai.pdf

E017 / P004 Additional end-to-end latency

Field	Evidence
Value	338.25
Units	ms
Baseline	Ready-model latency subtracted
Hardware	NVIDIA T416GB
Workload	Bert_base inference batch8 + training batch32
Interpretation	Training task occupies GPU memory and continues. Baseline condition matters; not a universal MPS comparison.
Evidence location	PDF pp11-12 / printed508-509; Table3; section6.1
Source	https://www.usenix.org/system/files/osdi20-bai.pdf

E018 / P005 Overall throughput ratio

Field	Evidence
Value	1.60
Units	x
Baseline	RT-Only dedicated GPU
Hardware	AMD MI5016GB
Workload	DISB A:1 VGG19 RT client100req/s uniform +1 ResNet152 BE
Interpretation	Includes RT and BE requests; RT-only reference dedicates GPU to RT tasks.
Evidence location	PDF pp11-12 / printed548-549; Table2; Figure11; section7.2
Source	https://www.usenix.org/system/files/osdi22-han.pdf

E019 / P005 RT end-to-end latency overhead

Field	Evidence
Value	0.5
Units	percent
Baseline	RT-Only dedicated GPU
Hardware	AMD MI5016GB

Field	Evidence
Workload	DISB A:1 VGG19 RT client100req/s uniform +1 ResNet152 BE
Interpretation	Described as negligible; not a p99 value or worst-case guarantee.
Evidence location	PDF pp11-12 / printed548-549; Table2; Figure11; section7.2
Source	https://www.usenix.org/system/files/osdi22-han.pdf

E020 / P005 Overall throughput ratio

Field	Evidence
Value	1.14
Units	x
Baseline	RT-Only dedicated GPU
Hardware	AMD MI5016GB
Workload	DISB B:1 VGG19 RT client220req/s uniform +1 ResNet152 BE
Interpretation	High RT load; differs from low-load DISB A.
Evidence location	PDF pp11-12 / printed548-549; Table2; Figure11; section7.2
Source	https://www.usenix.org/system/files/osdi22-han.pdf

E021 / P005 RT end-to-end latency overhead

Field	Evidence
Value	1
Units	percent
Baseline	RT-Only dedicated GPU
Hardware	AMD MI5016GB
Workload	DISB B:1 VGG19 RT client220req/s uniform +1 ResNet152 BE
Interpretation	About60 microseconds in prose; not a tail percentile.
Evidence location	PDF pp11-12 / printed548-549; Table2; Figure11; section7.2

Field	Evidence
Source	https://www.usenix.org/system/files/osdi22-han.pdf

E022 / P005 RT end-to-end latency overhead

Field	Evidence
Value	<1.5
Units	percent
Baseline	RT-Only dedicated GPU
Hardware	AMD MI5016GB
Workload	DISB C-E:multiple BE clients
Interpretation	Authors also state0.1ms; keep inequality and scenario range.
Evidence location	PDF pp11-12 / printed548-549; Table2; Figure11; section7.2
Source	https://www.usenix.org/system/files/osdi22-han.pdf

E023 / P005 Overall throughput ratio

Field	Evidence
Value	3.00
Units	x
Baseline	RT-Only dedicated GPU
Hardware	AMD MI5016GB
Workload	DISB D:5 RT model clients20req/s uniform +5 BE model clients
Interpretation	About25% lower throughput than GPUStreams; latency trade-off preserved.
Evidence location	PDF pp11-12 / printed548-549; Table2; Figure11; section7.2
Source	https://www.usenix.org/system/files/osdi22-han.pdf

E024 / P005 Overall throughput ratio

Field	Evidence
Value	2.96
Units	x
Baseline	RT-Only dedicated GPU
Hardware	AMD MI5016GB
Workload	DISB E:5 RT model clients20req/s Poisson +5 BE model clients
Interpretation	About25% lower throughput than GPUStreams; latency trade-off preserved.
Evidence location	PDF pp11-12 / printed548-549; Table2; Figure11; section7.2
Source	https://www.usenix.org/system/files/osdi22-han.pdf

E025 / P005 Overall throughput ratio

Field	Evidence
Value	7.7
Units	x
Baseline	RT-Only dedicated GPU
Hardware	AMD MI5016GB
Workload	Apollo REAL trace about43 RTreq/s +5 BE model clients
Interpretation	Real-world trace mapped to representative DNN models; not production deployment measurement.
Evidence location	PDF pp11-12 / printed548-549; Table2; Figure11; section7.2
Source	https://www.usenix.org/system/files/osdi22-han.pdf

E026 / P005 RT end-to-end latency overhead

Field	Evidence
Value	<2
Units	percent
Baseline	RT-Only dedicated GPU
Hardware	AMD MI5016GB

Field	Evidence
Workload	Apollo REAL trace about43 RTreq/s +5 BE model clients
Interpretation	Not a p99 latency claim or security/fault isolation test.
Evidence location	PDF pp11-12 / printed548-549; Table2; Figure11; section7.2
Source	https://www.usenix.org/system/files/osdi22-han.pdf

E027 / P006 ResNet18 p99 reduction factor

Field	Evidence
Value	up to3
Units	x
Baseline	CUDA-MS
Hardware	Tesla T4
Workload	`ResNet18+InceptionV3;lognormal` sigma1.5/2
Interpretation	Author-reported approximate maximum; not uniform across loads
Evidence location	PDF p10 / printed604; section7.1; Figure12 onPDFp11
Source	https://vincen.tl/files/ng23paella.pdf

E028 / P006 p90 instrumentation overhead

Field	Evidence
Value	6.6
Units	microseconds
Baseline	Uninstrumented kernel
Hardware	Tesla T4
Workload	Empty kernel;160 blocks;instrumentation with aggregation
Interpretation	Microbenchmark; not inference request latency
Evidence location	PDF p12 / printed606; section7.3; Figure15
Source	https://vincen.tl/files/ng23paella.pdf

E029 / P007 Aggregate throughput ratio

Field	Evidence
Value	up to7.3
Units	x
Baseline	Dedicated high-priority inference
Hardware	V10016GB;A10040GB extension
Workload	Two inference clients;Poisson;V100
Interpretation	Maximum; includes best-effort work
Evidence location	PDF p12 / printed1086; section6.2.3; Figure12
Source	https://fotstrt.github.io/files/2024-orion.pdf

E030 / P007 High-priority p99 overhead

Field	Evidence
Value	15
Units	percent
Baseline	Dedicated high-priority inference
Hardware	V10016GB;A10040GB extension
Workload	Two inference clients;Poisson;V100
Interpretation	Authors summarize within15%; averaged across collocations, not worst-case guarantee
Evidence location	PDF p12 / printed1086; section6.2.3; Figure12
Source	https://fotstrt.github.io/files/2024-orion.pdf

E031 / P008 GPU utilization efficacy GUE

Field	Evidence
Value	94.04
Units	composite score
Baseline	Default MPS GUE38.82

Field	Evidence
Hardware	One V10016GB
Workload	Alexnet+ResNet50;25ms target
Interpretation	Not percent GPU utilization
Evidence location	GSLICE PDF p9; printed p500; Table3
Source	https://www.cse.iitb.ac.in/approximately puru/courses/spring2024-25/downloads/gslice.pdf

E032 / P008 Signed latency improvement

Field	Evidence
Value	-7.52
Units	percent
Baseline	TensorRT server19.02
Hardware	One V10016GB
Workload	`ResNet50;batch1;GPU-ready` batch execution
Interpretation	Negative means worse latency; excludes transport overhead; single-model comparison
Evidence location	GSLICE PDF p10; printed p501; Table4 and section4.3
Source	https://www.cse.iitb.ac.in/approximately puru/courses/spring2024-25/downloads/gslice.pdf

E033 / P009 Clockwork goodput

Field	Evidence
Value	approximately800
Units	requests/second
Baseline	INFaaS on nonidentical EC2 setup
Hardware	One V10032GB in section6.1
Workload	15 ResNet50 instances;16 clients/model;500ms SLO
Interpretation	Within-SLO successes; comparator qualitative; not total throughput

Field	Evidence
Evidence location	USENIX PDF pp10-11; printed pp451-452; Figure5 and section6.1
Source	https://people.mpi-sws.org/approximately antoinek/documents/20osdi_clockwork.pdf

E034 / P009 Clockwork request latency

Field	Evidence
Value	approximately300
Units	milliseconds
Baseline	No comparative ratio
Hardware	One V10032GB in section6.1
Workload	15 ResNet50 instances;16 clients/model;500ms SLO
Interpretation	Batch16 scheduling explanation; not a p99 value
Evidence location	USENIX PDF pp10-11; printed pp451-452; Figure5 and section6.1
Source	https://people.mpi-sws.org/approximately antoinek/documents/20osdi_clockwork.pdf

E035 / P010 Average P99 inference latency improvement factor, as reported

Field	Evidence
Value	12.6
Units	x
Baseline	TaskSwitch (authors' adapted temporal-switch baseline)
Hardware	single NVIDIA Tesla V100 16GB in section6.1; four-V100 host; separate section6.6 single A100 80GB LLM experiment
Workload	section6.1 single-V100 DNN inference+Swin-T; `Light/Heavy/Burst/Skewed/MAF` traces;300s
Interpretation	Author-reported average across tested workloads; not a uniform per-model speedup, and not a comparison to unmodified PipeSwitch.

Field	Evidence
Evidence location	USENIX PDF p11; printed p1666; Figure10 discussion
Source	https://www.usenix.org/system/files/atc25-wang-jiali.pdf

E036 / P011 Reported throughput

Field	Evidence
Value	446
Units	requests/s
Baseline	same configuration with1 process (354requests/s)
Hardware	A10080GB; Amazon p4de.24xlarge hosts with8 GPUs; extracted profiler point concerns one MIG segment
Workload	InceptionV3,1-GPC MIG instance,batch4;3 MPS processes
Interpretation	One local configuration, not fleet savings. Source is arXivv1; final published version not verified.
Evidence location	arXiv v1 PDF p5; sectionIII-B; Figures3-4 on p4
Source	https://arxiv.org/pdf/2409.14447

E037 / P011 Reported inference latency; quantile unspecified

Field	Evidence
Value	27
Units	ms
Baseline	same configuration with1 process (11ms)
Hardware	A10080GB; Amazon p4de.24xlarge hosts with8 GPUs; extracted profiler point concerns one MIG segment
Workload	InceptionV3,1-GPC MIG instance,batch4;3 MPS processes
Interpretation	Quantile/arrival model not established for this profiler point; do not label p99. Value is specific to the accessed author manuscript.
Evidence location	arXiv v1 PDF p5; sectionIII-B; Figures3-4 on p4
Source	https://arxiv.org/pdf/2409.14447

Field	Evidence

E038 / P012 Reported single-GPU throughput factor

Field	Evidence
Value	>10
Units	x
Baseline	Native early binding at same offered function rate
Hardware	single V10032GB in local Figure8; evaluation worker has4 V10032GB GPUs
Workload	multiple ResNet152 functions on one `V10032GB;10requests/minute/function`
Interpretation	Lower bound reported by authors, not an exact10x; function count varies to saturate memory. No fleet cost claim.
Evidence location	USENIX PDF pp10-11; printed pp605-606; Figure8 and section7.2
Source	https://www.usenix.org/system/files/atc25-yu.pdf

E039 / P012 Reported P98 latency upper bound

Field	Evidence
Value	<50
Units	ms
Baseline	Native under matched request-rate setting
Hardware	single V10032GB in local Figure8; evaluation worker has4 V10032GB GPUs
Workload	same single-GPU low-rate ResNet152 `test,10-80requests/minute/function`
Interpretation	Bound from text, not a digitized point; default98th percentile from section7 metrics. Applies to this workload.
Evidence location	USENIX PDF pp10-11; printed pp605-606; default P98 metrics and section7.2
Source	https://www.usenix.org/system/files/atc25-yu.pdf

E040 / P013 Author-reported average throughput factor

Field	Evidence
Value	6.7
Units	x
Baseline	INFless with documented warm-up/pipelined loading/model sharing
Hardware	V10016GB,80CUs; host has4 GPUs; single-GPU multi-user test in section9.4
Workload	three workflow `types,PyTorch/TensorFlow,Azure-derived` invocation patterns; evaluated V100 setup
Interpretation	Aggregate within-paper result across tested workflows; not universal or a matched cross-paper speedup. Batch/concurrency optimizations contribute.
Evidence location	USENIX PDF p11; printed p68; section9.1; Figure13
Source	https://www.usenix.org/system/files/atc24-wu-hao.pdf

E041 / P013 Reported API-forwarding overhead range

Field	Evidence
Value	3-5
Units	microseconds
Baseline	direct API path; instrumentation overhead measurement
Hardware	V10016GB,80CUs; host has4 GPUs; single-GPU multi-user test in section9.4
Workload	GPU API forwarding with48concurrent functions on one GPU
Interpretation	Per-API overhead, not end-to-end inference latency; many APIs asynchronous.
Evidence location	USENIX PDF p13; printed p70; section9.3; Figure18
Source	https://www.usenix.org/system/files/atc24-wu-hao.pdf

E042 / P014 Reported average QoS violation rate

Field	Evidence
Value	1.23
Units	percent of requests
Baseline	original soloP99 threshold (1%baseline violations by definition)
Hardware	A10040GB for local quota/QoS tests; twoRTX4090 used for separate distributed-training test
Workload	Transformer-XL inference; batch8/16/32. Original rate is half solo maximum; shared test halves that rate again and assigns 50% time; tokens5-80ms.
Interpretation	Average across tested configurations; no claim that worst-case violation is1.23%. Threshold-relative metric, not latency in milliseconds.
Evidence location	USENIX PDF pp12-13; printed pp1013-1014; section5.5; Figure17
Source	https://www.usenix.org/system/files/atc25-zhang-shulai.pdf

E043 / P014 Reported kernel module host-memory footprint

Field	Evidence
Value	4.8
Units	MB
Baseline	no module; additional host memory cost
Hardware	A10040GB for local quota/QoS tests; twoRTX4090 used for separate distributed-training test
Workload	kernel-space GPU sharing module on evaluated Linux host
Interpretation	Host memory, not GPU VRAM; separate monitoring thread also required.
Evidence location	USENIX PDF p13; printed p1014; section5.7
Source	https://www.usenix.org/system/files/atc25-zhang-shulai.pdf

E044 / P015 Reported high-priorityP99 latency ratio

Field	Evidence
Value	1.07
Units	x standalone
Baseline	standalone high-priorityBert-large execution
Hardware	single NVIDIA GV100 for extracted Triton/Paella case; NPU/ASIC/FPGA results outside extracted GPU outcomes
Workload	twoBert-large clients `ononeGV100;high-priority10req/s,background continuous`
Interpretation	Triton+XSched;not a rawmillisecond value. NativeTriton `ratio1.53x;noNPUresultincluded.`
Evidence location	PDF p14; printed p683; section8 case3; Figure15a
Source	https://www.usenix.org/system/files/osdi25-shen-weihang.pdf

E045 / P015 Author-reported P99 latency reduction

Field	Evidence
Value	30.0
Units	percent
Baseline	vanillaTriton (P99=1.53xstandalone)
Hardware	single NVIDIA GV100 for extracted Triton/Paella case; NPU/ASIC/FPGA results outside extracted GPU outcomes
Workload	same `two-clientTriton/GV100case`
Interpretation	Within-paper comparison only;author-rounded30.0%,not a universal bound.
Evidence location	PDF p14; printed p683; section8 case3; Figure15a
Source	https://www.usenix.org/system/files/osdi25-shen-weihang.pdf

E046 / P018 SLO-compliant goodput

Field	Evidence
Value	524.29

Field	Evidence
Units	requests/s
Baseline	Clockwork: 457.24 requests/s
Hardware	Dell PowerEdgeR740; V10032GB; local synthetic co-running experiment and separate four-GPU realistic trace.
Workload	Local three-model load ratio 3:1:2
Interpretation	Local outcome, not four-GPU replay or p99
Evidence location	PDFp9 / `publishedp18;Table3;unfairness` definitionPDFp2 / publishedp11;Eq2
Source	https://www.sciencedirect.com/science/article/pii/S0167739X23003187

E047 / P018 Unfairness ratio

Field	Evidence
Value	1.05
Units	ratio
Baseline	Clockwork: 1.50
Hardware	Dell PowerEdgeR740; V10032GB; local synthetic co-running experiment and separate four-GPU realistic trace.
Workload	Local three-model load ratio 3:1:2
Interpretation	Maximum/minimum model slowdown; not security isolation
Evidence location	PDFp9 / `publishedp18;Table3;unfairness` definitionPDFp2 / publishedp11;Eq2
Source	https://www.sciencedirect.com/science/article/pii/S0167739X23003187

E048 / P019 Pairs meeting twice-solo p99 QoS target

Field	Evidence
Value	30
Units	tested pairs
Baseline	MPS Non-ArkGPU

Field	Evidence
Hardware	Machine with four V100s; Xeon Platinum8163; CentOS7.9.2009; driver460.106.00; CUDA11.1; cuDNN7.6.5.
Workload	Five inference models paired with six training models
Interpretation	Count of tested pairs, not a universal guarantee
Evidence location	PDF p11 / printed p314; section5.2; Figure14 on PDFp12
Source	https://doi.org/10.1007/s42514-023-00154-y

E049 / P019 Low-contention QPS loss

Field	Evidence
Value	about 5
Units	percent
Baseline	Unthrottled co-location
Hardware	Machine with four V100s; Xeon Platinum8163; CentOS7.9.2009; driver460.106.00; CUDA11.1; cuDNN7.6.5.
Workload	Low-contention tested pair settings
Interpretation	Regression retained; larger system-level goodput gains are not one-GPU results
Evidence location	PDF p12 / printed p315; section5.2; Figure16 on PDFp13
Source	https://doi.org/10.1007/s42514-023-00154-y

E050 / P020 Effect of 15 percent prediction perturbation

Field	Evidence
Value	8.09
Units	percent makespan increase
Baseline	Unperturbed predictor
Hardware	Two A30s,24GB each; CUDA12.2; driver535.171.04; DCGM3.1.3. Local sections4.2/4.3.1 use one A30. StaCon `profiles2g.12gb+1g.6gb+1g.6gb`.
Workload	Reported prediction sensitivity

Field	Evidence
Interpretation	Sensitivity result; local JCT and cluster simulation are separate
Evidence location	PDF pp13-14; section4.2; Figure12 on PDFp16
Source	https://doi.org/10.1007/s10586-025-05284-2

E051 / P021 Evaluated latency statistic

Field	Evidence
Value	95
Units	percentile
Baseline	AMD unrestricted concurrency labelled MPSDefault
Hardware	One MI50/60CUs; dualEPYC7302/512GB; Ubuntu18.04; kernel5.4; ROCm4.5
Workload	Inference workers and heterogeneous pairs
Interpretation	Statistic definition, not a 95-percent latency improvement
Evidence location	Author PDF pp9-11; sectionVI; TableIII and Figure13b
Source	https://danielwong.org/files/KRISP-HPCA2023.pdf

E052 / P022 Normalized total system throughput

Field	Evidence
Value	1.19
Units	ratio
Baseline	TGS: 1.65
Hardware	Shared A100 SXM 40GB; eight-GPU EC2 p4d host inventory is not an eight-GPU outcome; `Ubuntu20.04/CUDA12.2/PyTorch2.2.0/ONNX` Runtime GPU1.17.0/Hidet0.3.0
Workload	Ten percent idle-time setting
Interpretation	Sum of normalized throughput of all concurrent workloads: Tally1.19 versus TGS1.65 at10% idle time. Not background throughput alone.

Field	Evidence
Evidence location	arXiv v3 PDF p10; section5.3; Figure6a on p11
Source	https://arxiv.org/pdf/2410.07381v3

D Calculated quantities

Calculation	Formula and input	Value / limit
C001 P011	(446/354-1)*100: 446requests/s at3processes;354requests/s at1process;InceptionV3,1-GPC,batch4	25.988700564971758 percent. calculated,not reported;based on accessed author manuscript
C002 P011	27/11: 27ms at3processes;11ms at1process;same1-GPC/batch4configuration	2.4545454545454546 x. calculated,not reported;quantile unspecified;based on accessed author manuscript

E Guideline deliverable crosswalk

Item	Requirement	Artifact	Status / qualification
1	Final SLR report	report/main.pdf (draft)	Draft present; verify before final submission
2	Engineering problem	report section 1	Draft present; verify before final submission
3	Research objectives	report section 1	Draft present; verify before final submission
4	Research questions	report section 2	Draft present; verify before final submission
5	SLR protocol	../task1/submitted_protocol.pdf	Existing evidence preserved; limits documented

Item	Requirement	Artifact	Status / qualification
6	Database justification	report section 3	Draft present; verify before final submission
7	Complete search strings	Search Log / appendix	Draft present; verify before final submission
8	Inclusion/exclusion criteria	report section 3	Draft present; verify before final submission
9	Database-wise results	report section 4	Existing evidence preserved; limits documented
10	Search and screening log	Screening Log / Search Log	Existing evidence preserved; limits documented
11	Duplicate record	Duplicate Removal	Existing evidence preserved; limits documented
12	PRISMA flow	`baseline/PRISMA_2020_FLOW_DIAGRAM.pdf`	Draft present; verify before final submission
13	Quality checklist/scores	QA / QA Reconciliation	Existing evidence preserved; limits documented
14	Included primary studies	Studies / Source Register	Draft present; verify before final submission
15	Extraction sheet	Extraction / Evidence	Existing evidence preserved; limits documented
16	Taxonomy	Taxonomy / report section 5	Draft present; verify before final submission
17	Comparative matrix	Comparability / report section 7	Draft present; verify before final submission

Item	Requirement	Artifact	Status / qualification
18	Trade-offs	report section 8	Draft present; verify before final submission
19	Conflicting evidence	Comparability / report section 9	Draft present; verify before final submission
20	All RQ answers	report section 6	Draft present; verify before final submission
21	Engineering recommendation	Recommendations / report section 10	Draft present; verify before final submission
22	Research gaps	Research Gaps / report section 11	Draft present; verify before final submission
23	Threats to validity	Validity / report section 12	Draft present; verify before final submission
24	References	report references	Draft present; verify before final submission
25	Supporting verification files	logs / baseline / scripts	Draft present; verify before final submission

F Journal benchmark and drafting standard

The organizational benchmark is Hong et al. [35]. Its layer-based taxonomy and comparison structure inform the draft. The current review adds the instructor-required search, selection and appraisal records. Journal standing is contextual and does not certify the draft as publishable. See the separate benchmark note for the dated publisher-metric check.

G Files, timestamps and remaining checks

The frozen report, workbook, ZIP and submitted protocol remain separate immutable artifacts. This dated working report integrates the seven confirmed additions. Original timestamps retain their historical meaning. The electronic CSV registers carry the full559screening decisions,325open advanced records, eligibility evidence, access attempts

and methodology corrections. The priority is to finish eligibility/identity closure, then reconcile whole-review PRISMA and regenerate a consistently dated submission package. No reviewer name or vote is inferred.

H Seven added source-based study packages

H.1 S021 / P024 LithOS

[9]

Field	Source-based record
Record ID	R00080
Title	LithOS: An Operating System for Efficient Machine Learning on GPUs
Venue/year	Proceedings of the ACM SIGOPS 31st Symposium on Operating Systems Principles; 2025
DOI	10.1145/3731569.3764818
Source	https://www.pdl.cmu.edu/PDL-FTP/BigLearning/lithos_sosp25.pdf
Version evidence hash	92cc03bf90bcacf38862d1df5d3630545b1502b71abe4aae15d219f06445f873
Eligibility	Include; all IC1-IC6 met on inspected primary text
Decision date UTC	2026-10-22T22:37:00.954370+00:00

Criterion	Evidence and anchor
IC1	SOSP 2025 typeset English primary conference paper, DOI and 17-page citation on first page. PDF p1
IC2	TPC allocation, kernel atomization and CUDA-driver interposition schedule independent GPU contexts; implementation builds on MPS. PDF pp6-10; sections5-6
IC3	Inference-only setup has two high-priority model applications and one best-effort inference app; mixed training experiment is separate. PDF pp10-11; section7, Table2 and section8.1
IC4	SLO attainment, normalized throughput/goodput and end-to-end p99 latency are defined and compared under stated load. PDF pp11-12; section8.1, Figures14-16
IC5	Primary testbed is one A100 SXM4, 40GB, 108SM; local inference app combinations are evaluated. PDF pp10-11; section7 and section8.1
IC6	2025 final publication is within the submitted 2020-to-final-search window and predates the recorded October2026 search checkpoint. PDF p1; submitted protocol section6

QA	Score	Rationale and source anchor
Q1	2/2	Scheduling, efficiency and isolation objectives are explicit. p1; section1
Q2	2/2	TPC scheduling, atomization, right-sizing and driver interposition are described. pp6-10; sections5-6
Q3	1/2	Setup and outcome provenance are reported; repeated end-to-end trials and uncertainty need confirmation. pp10-13; sections7-8
Q4	2/2	One A10040GB, software stack, model families and arrivals are specified. pp10-11; Table2 and section7
Q5	1/2	Comparisons disclose reimplementation, unequal partition fractions and absence of BE for MIG/Limits; identical tenant capacity is not maintained. pp10-11; sections7-8.1
Q6	2/2	Normalized throughput, app goodput and SLO-attainment axes are defined. p11; Figures14-15
Q7	2/2	End-to-end p99, per-model constraints and request rates are stated. pp10-12; Table2, Figure16
Q8	2/2	Multiple inference combinations and hybrid inference/training co-tenancy are tested. pp11-12; section8.1
Q9	2/2	Performance-isolation boundary is tested through HP SLO attainment and BE progress; this is not a security/fault guarantee. p11; Figures14-15
Q10	1/2	Implementation details and special-kernel limits aid replication; complete artifact execution has not been verified. p10; section6

LITH-01 SLO attainment

Field	Extracted value/context
Metric	SLO attainment
Value	100
Unit	percent
Baseline	MPS45percent; TGS84percent in same plotted scenario
Context	One A100; twoHP plusBE except MIG/Limits omitBE; model combinations and loads in Table2
Anchor	p11 section8.1 Figure14
Interpretation limit	Scenario result, not universal hardware guarantee

LITH-02 Normalized aggregate throughput

Field	Extracted value/context
Metric	Normalized aggregate throughput
Value	1
Unit	normalized
Baseline	MPS1.11; MIG0.71; Limits0.58
Context	Same scenario; normalized definition and unequal supported tenant sets retained
Anchor	p11 section8.1 Figure14
Interpretation limit	Compare benefit with SLO attainment and BE availability

H.2 S022 / P025 Abacus

[10]

Field	Source-based record
Record ID	R00143
Title	Enable simultaneous DNN services based on deterministic operator overlap and precise latency prediction
Venue/year	Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis; 2021
DOI	10.1145/3458817.3476143
Source	https://www.microsoft.com/en-us/research/wp-content/uploads/2022/07/3458817.3476143.pdf
Version evidence hash	5971fe513e1b16a3134574c7dba37930074230654ed0aba63c43c9163ce0a876
Eligibility	Include; all IC1-IC6 met on inspected primary text
Decision date UTC	2026-10-22T22:37:00.954370+00:00

Criterion	Evidence and anchor
IC1	SC2021 typeset English primary conference paper identifies DOI, venue and 14-page main paper; supplied PDF also has an artifact appendix. PDF p1 and p15
IC2	Separate DNN service processes are coordinated through segmental executors and deterministic overlapped operator groups on a shared GPU. PDF pp7-9; sections6.1-7.1
IC3	Seven vision/NLP inference services are tested in pairwise and higher-order co-location. PDF pp6,9-11; Table1 and sections7.1-7.4
IC4	p99 latency, QoS violation and queries-per-second throughput are defined with load and baseline policies. PDF pp9-10; sections7.1-7.3 and Figures14-17
IC5	Local experiments run on one A100 with MPS; a separate 16-GPU cluster experiment is distinguishable. PDF pp9,11; sections7.1,7.5-7.6
IC6	2021 conference publication is within the submitted main window beginning1January2020. PDF p1; submitted protocol section6

QA	Score	Rationale and source anchor
Q1	2/2	Objective is predictable concurrent service latency with higher throughput. p1
Q2	2/2	Operator-group prediction, segmental executors and headroom search are described. pp4-8; sections4-6
Q3	1/2	Load, sampling and measurements are detailed; serving-run repeat variability remains to be verified. pp6-10; sections5.4,7.1-7.3
Q4	1/2	Software, models and arrivals are documented but Table2 reports an unresolved A100128SM description. p9 Table2; p15 artifact description
Q5	2/2	FCFS/SJF/EDF baselines use matched workload generation; query-drop treatment is disclosed. p9 section7.1
Q6	2/2	Peak throughput is explicitly queries/s, distinguished from images/sequences per second. p10 section7.3
Q7	2/2	p99 and QoS targets are defined with50QPS and separate100QPS peak experiment. pp9-10; sections7.1-7.3
Q8	2/2	Pairwise21combinations plus triple/quadruple co-location are tested. pp9-11 sections7.2-7.4
Q9	1/2	Performance interference is measured, but separate-process privacy/crash motivation is not a demonstrated GPU security/fault-isolation guarantee. pp7,9-11; sections6.1,7
Q10	2/2	Artifact appendix and profiling/scheduling overhead discussion provide replication support and limits. pp12,15; section7.8 and artifact appendix

ABA-01 Mean p99 latency reduction

Field	Extracted value/context
Metric	Mean p99 latency reduction
Value	23.1;34.1;23.8
Unit	percent
Baseline	FCFS;SJF;EDF respectively
Context	21pairwise model combinations at50QPS on A100/MPS; latency excludes baseline dropped queries
Anchor	p9 section7.2 Figure14

Field	Extracted value/context
Interpretation limit	QoS violation metric separately counts dropped requests

ABA-02 Mean peak throughput improvement

Field	Extracted value/context
Metric	Mean peak throughput improvement
Value	25.7;38.1;25.7
Unit	percent
Baseline	FCFS;SJF;EDF respectively
Context	100QPS offered load; throughput is queries/s, not images/s
Anchor	p10 section7.3 Figure17
Interpretation limit	VGG16/VGG19 slight degradation retained as adverse result

H.3 S023 / P026 Miriam

[11]

Field	Source-based record
Record ID	R01089
Title	Miriam: Exploiting Elastic Kernels for Real-time Multi-DNN Inference on Edge GPU
Venue/year	Proceedings of the 21st ACM Conference on Embedded Networked Sensor Systems; 2023
DOI	10.1145/3625687.3625789
Source	https://neawhen.github.io/neiwen.github.io/assets/pdf/sensys23-miriam.pdf
Version evidence hash	43fd9943515edcdb87632ee89004c408ccac1063c58323a5614b48cb228f6d46
Eligibility	Include; all IC1-IC6 met on inspected primary text
Decision date UTC	2026-10-22T22:37:00.954370+00:00

Criterion	Evidence and anchor
IC1	English SenSys2023 typeset14-page primary paper with exact DOI; frozen2024 year conflicts with first-page2023 publication and requires metadata correction. PDF p1
IC2	Elastic kernel transformation and runtime coordination control co-execution between critical and normal DNNs. PDF pp5-8; sections5-8
IC3	Mixed-critical DNN Task Benchmark explicitly uses concurrent critical and normal inference models. PDF p9; section9.1.2 and Table2
IC4	End-to-end critical latency, overall request throughput and achieved occupancy are defined and evaluated. PDF pp9-11; sections9.1.4-9.3
IC5	Per-device experiments compare co-running workloads on RTX2060 and an edge platform; platform-label inconsistency must remain flagged. PDF p9; section9.1 and Table2
IC6	Typeset SenSys2023 publication is inside2020-to-final-search window; no historical exception is needed. PDF p1; submitted protocol section6

QA	Score	Rationale and source anchor
Q1	2/2	Mixed-criticality inference latency/throughput objective is explicit. pp1-2
Q2	2/2	Elastic kernel generator and runtime coordination are explained. pp5-8; sections5-8
Q3	1/2	Workloads and comparison metrics are described; repeated-run uncertainty is not established in inspected evaluation. pp9-11; section9
Q4	1/2	Models, arrival patterns and CUDA stack are reported; edge GPU name/architecture/core count is internally inconsistent. pp2,9; section9.1
Q5	2/2	Three baselines and operator-fusion settings are specified with reasons for exclusions. p9 section9.1.3
Q6	2/2	Overall request throughput and achieved occupancy are defined. p9 section9.1.4; p10 Figure9
Q7	1/2	Critical-task end-to-end latency is plotted, but run interval and quantile/repetition interpretation need confirmation. pp9-10 Figure9
Q8	2/2	Critical and normal task pairs are evaluated under closed-loop, uniform and Poisson patterns. pp9-10 Table2 and Figure9
Q9	1/2	Performance protection is measured but under10percent introductory summary does not cover all reported overheads; no security/fault guarantee is established. pp1,9-10 Figure9
Q10	2/2	Source-code link, baseline details and compiler/runtime limitations aid replication. pp2,9; sections8-9

MIR-01 Critical latency ratio

Field	Extracted value/context
Metric	Critical latency ratio
Value	1.21;1.28
Unit	normalized to Sequential
Baseline	Sequential1.0
Context	MDTB-A closed-loop AlexNet/CifarNet; RTX2060 and platform labeledXavier respectively
Anchor	p10 Figure9a-b

Field	Extracted value/context
Interpretation limit	21percent and28percent overhead; contradicts universal under10percent reading

MIR-02 Overall throughput ratio

Field	Extracted value/context
Metric	Overall throughput ratio
Value	1.64;1.83
Unit	normalized to Sequential
Baseline	Sequential1.0
Context	MDTB-A; RTX2060 and platform labeledXavier respectively
Anchor	p10 Figure9c-d
Interpretation limit	Do not pair maximum throughput from a different workload with these latency values

H.4 S024 / P027 SGDRC

[12]

Field	Source-based record
Record ID	R00656
Title	SGDRC: Software-Defined Dynamic Resource Control for Concurrent DNN Inference on NVIDIA GPUs
Venue/year	Proceedings of the 30th ACM SIGPLAN Annual Symposium on Principles and Practice of Parallel Programming; 2025
DOI	10.1145/3710848.3710863
Source	https://huaicheng.github.io/p/ppopp25-sgdrc.pdf
Version evidence hash	dd206161e532cecc02afc7ceb1b25f4980cfef59342575570548 62094866cb58
Eligibility	Include; all IC1-IC6 met on inspected primary text
Decision date UTC	2026-10-22T22:37:00.954370+00:00

Criterion	Evidence and anchor
IC1	English typeset PPoPP2025 primary paper identifies the exact DOI and conference publication. PDF p1
IC2	Software page coloring, bimodal tensors and tidal SM masks dynamically allocate memory bandwidth and compute between services. PDF sections5-8
IC3	Latency-sensitive inference services are colocated with best-effort inference models; workloads and arrival trace are specified. PDF p10; Table3 and section9.2
IC4	p99 including queueing, SLO attainment, best-effort sample throughput and overall goodput/throughput are defined. PDF pp10-11; sections9.2-9.3
IC5	P40 and RTX A2000 are evaluated as local devices with clients/server on the same host; local co-tenant outcomes are inspectable. PDF p10 section9.2
IC6	March2025 conference publication predates recorded October2026 search and is in the main window. PDF p1; submitted protocol section6

QA	Score	Rationale and source anchor
Q1	2/2	Responsiveness and throughput under inference co-location are explicit goals. pp1-2
Q2	2/2	Page coloring, bimodal tensors and dynamic SM masks define the resource-control mechanism. sections5-8
Q3	1/2	Trace scaling and metrics are specified; serving-run repetitions and uncertainty remain unconfirmed. p10 section9.2
Q4	1/2	Devices, model list and arrivals are given, but text says12models while Table3 lists11. p10 Table3 and section9.2
Q5	1/2	Baseline modifications and fixed MPS split are disclosed; Orion is a policy reimplementation, limiting equivalence. p10 section9.2
Q6	2/2	Overall throughput combines LS goodput and BE samples/s; per-BE throughput is tracked. pp10-11 section9.2 and Figure18
Q7	2/2	p99 includes queueing and SLO uses n times isolated p99; light/heavy trace conditions are stated. p10 section9.2
Q8	2/2	Concurrent LS services and BE model combinations are evaluated on two device types. pp10-11 Figure18
Q9	2/2	Performance allocation is tested and lack of fault isolation is explicitly acknowledged; no fault guarantee is inferred. p11 section10
Q10	1/2	Implementation and limits are detailed, but complete executable artifact reproduction has not been verified. p11 section10

SGD-01 Overall throughput ratio

Field	Extracted value/context
Metric	Overall throughput ratio
Value	up to1.47;up to1.32
Unit	times
Baseline	Orion policy reimplemented in SGDRC
Context	P40 and RTX A2000 respectively; co-located inference; light/heavy Apollo trace settings
Anchor	p11 section9.3 Figure18
Interpretation limit	Maxima are device-specific; MPS is unsupported on P40 in this evaluation

SGD-02 Best-effort throughput ratio

Field	Extracted value/context
Metric	Best-effort throughput ratio
Value	up to2.36;up to1.79
Unit	times
Baseline	Orion policy reimplemented in SGDRC
Context	P40 and RTX A2000 respectively; LS goodput must be assessed alongside BE throughput
Anchor	p11 section9.3 Figure18
Interpretation limit	Does not establish fault isolation; shared CUDA context cannot isolate runtime errors

H.5 S025 / P028 Pantheon

[13]

Field	Source-based record
Record ID	R00661
Title	Pantheon: Preemptible Multi-DNN Inference on Mobile Edge GPUs
Venue/year	Proceedings of the 22nd Annual International Conference on Mobile Systems, Applications and Services; 2024
DOI	10.1145/3643832.3661878
Source	https://zhouzimu.github.io/paper/mobisys24-han.pdf
Version evidence hash	00ebe79cb2fc3c8d49ff48ab38031e1e18116d196bbdfb858722c191ac442f44
Eligibility	Include; all IC1-IC6 met on inspected primary text
Decision date UTC	2026-10-22T22:37:00.954370+00:00

Criterion	Evidence and anchor
IC1	Final English MobiSys2024 primary14-page paper identifies exact DOI and conference on first page. PDF p1
IC2	Middleware schedules distinct concurrent DNN tasks with chunk-level preemption, model variants, and high/low-priority CUDA streams. Scheduler and worker are separate threads, not separate secure processes. PDF pp3-8; sections3-4
IC3	Nine real-time inference task types from smart traffic, service robot and UAV applications run with an additional best-effort AlexNet. PDF p8; Table1 and section4
IC4	Deadline miss rate and deadline-adjusted relative accuracy are defined; GPU utilization, best-effort interference and scheduling overhead are evaluated. PDF pp8-11; sections5.1-5.5
IC5	Main experiments use one Jetson Xavier NX GPU; a separate Nano deployment and NX car case study are identifiable local-device experiments. PDF pp7-8,10; sections4 and5.3
IC6	June2024 MobiSys publication is inside the original2020-to-search window. PDF p1

QA	Score	Rationale and source anchor
Q1	2/2	Objective is timely concurrent DNN inference with acceptable accuracy and memory cost. pp1-3
Q2	2/2	Chunk slicing, preemption, nested redundancy, variant selection and runtime architecture are described. pp3-8; sections3-4
Q3	1/2	Arrival jitter, loads and some variance results are reported; broad repeated-trial and uncertainty reporting is incomplete. pp8-10; sections4-5.3
Q4	1/2	Hardware and task table are detailed, but service-robot/UAV model labels differ between Table1 and following prose. pp7-8; Table1 and application tasks
Q5	2/2	RMS, DMS and RTm-DL are specified; RTm-DL uses the same accuracy constraint and all methods fix the GPU frequency. p8; section5
Q6	2/2	Utilization is compared under increasing concurrent tasks; values are contextual, not a universal efficiency gain. p9; Figure11
Q7	2/2	Deadline miss definition, task deadlines, completion-time traces and nonperiodic arrivals are provided. pp8-10; Table1, Figures9-10,15
Q8	2/2	Best-effort arrival load and increasing real-time task counts test co-tenant interference. pp9-10; sections5.1,5.3
Q9	1/2	Stream-priority and scheduling isolation are described. Runtime uses shared memory/threads; no GPU fault or adversarial isolation guarantee is established. pp7-8; section4; p10 best-effort load
Q10	1/2	Implementation stack, profiling limits and memory trade-offs are described; full reproducibility/calibration is not established by this review. pp7-8,11-12; sections4,6

PANT-01 Deadline miss rate

Field	Extracted value/context
Metric	Deadline miss rate
Value	0.39-1.10
Unit	percent
Baseline	RMS; DMS; RTm-DL
Context	Three application mixes on Xavier NX; periodic30fps tasks plus best-effort AlexNet; variants constrain accuracy/memory

Field	Extracted value/context
Anchor	PDF p9 section5.1 Figure9
Interpretation limit	Observed deadline performance; early exits alter models and late jobs count as zero accuracy.

PANT-02 GPU utilization

Field	Extracted value/context
Metric	GPU utilization
Value	50.85-78.50
Unit	percent
Baseline	RTm-DL46.10-80.05percent
Context	Increasing service-robot task counts; Figure10-11 experiment
Anchor	PDF p9 section5.1 Figure11
Interpretation limit	Similar utilization accompanies lower deadline misses; no general utilization dominance.

PANT-03 Mean device power

Field	Extracted value/context
Metric	Mean device power
Value	10.15
Unit	W
Baseline	RTm-DL9.73W
Context	NX INA3221 power monitor; idle4.01W; added online scheduler computation
Anchor	PDF p11 section5.5 Figure18
Interpretation limit	Device power is not per-query energy; authors report4.3percent power increase.

H.6 S026 / P029 Oneiros

[14]

Field	Source-based record
Record ID	R00086
Title	Oneiros: KV Cache Optimization through Parameter Remapping for Multi-tenant LLM Serving
Venue/year	Proceedings of the 2025 ACM Symposium on Cloud Computing; 2025
DOI	10.1145/3772052.3772215
Source	https://lca.ece.utexas.edu/pubs/Li_SoCC_2025.pdf
Version evidence hash	ea781c29e211cb9269a7c66893ebbc617cefb76ad0ae3e507ab2f32298ab7f51
Eligibility	Include; all IC1-IC6 met on inspected primary text
Decision date UTC	2026-10-22T22:37:00.954370+00:00

Criterion	Evidence and anchor
IC1	English SoCC2025 final14-page primary conference paper; exact DOI and venue on p1; publisher proceeding independently confirms article identity. PDF p1; publisher10.1145/3772052
IC2	Parameter-memory remapping, controller and asynchronous transfer engine support temporal and spatial multi-model GPU sharing, including MPS and MIG scenarios. PDF sections4-6; pp9-11 section7.2-7.3
IC3	LLM inference workloads comprise distinct OPT and Llama model instances; C1 has three models and C2 has two. PDF p9 Table1 and section7.1
IC4	P99 TTFT/TBT and generated tokens/s are defined; per-configuration temporal and spatial comparisons are supplied. TTFT includes queuing. PDF pp9-11 sections7.1-7.4; p11 footnote6
IC5	Experiments use one GH200 system and concurrent local model combinations; CPU memory extends GPU capacity. Hardware labels require clarification. PDF p9 section7.1 and Table1
IC6	SoCC conference is November2025; the queue indexes2026. Both dates are within the submitted window. Preserve the distinction between venue year and publication/index date. PDF p1; frozen queueR00086

QA	Score	Rationale and source anchor
Q1	2/2	Memory pressure and latency/throughput objectives for multi-tenant serving are explicit. p1; section1
Q2	2/2	Memory remapping, layer selection, asynchronous engine and controller policies are described. sections4-6
Q3	1/2	Traces, datasets and load variations are reported; repeated-trial uncertainty and seeds are not established by this review. p9 section7.1; pp9-12 evaluation
Q4	1/2	Versions, platform and model reservations are supplied; GH200/H20096GB description and Figure13 isolation caption need clarification. p9 section7.1; p11 Figure13
Q5	2/2	vLLM and Pie baselines, runtime versions and same-platform comparisons are stated; outcome contexts retained. pp9-11 sections7.1-7.4
Q6	2/2	Tokens/s and memory-reservation contexts are reported for temporal and spatial sharing. p9 Table1; pp10-11 Figures8-14
Q7	2/2	P99 TTFT/TBT are defined; queuing inclusion and varying arrivals are explicit. p9 section7.1; p11 footnote6
Q8	2/2	C1/C2, unequal arrival rates and input-length combinations test interactions between models. pp9-10 Table1; Figures9-10
Q9	1/2	MPS and MIG boundaries are discussed, but isolation wording and Figure13 caption conflict; no fault/adversarial isolation test is demonstrated. pp9-11 section7.3; Figure13
Q10	2/2	Source repository, software versions and transfer/idle-load trade-offs and ablations are supplied. Artifact was not executed here. p1 repositoryURL; p9 section7.1; pp11-12 sections7.5-7.6

ONEI-01 P99 TBT reduction

Field	Extracted value/context
Metric	P99 TBT reduction
Value	65.5
Unit	percent
Baseline	vLLM
Context	C1 three-model combination; Alpaca/ShareGPT and Azure coding traces; author-reported average across varied arrival rates under non-strict spatial sharing

Field	Extracted value/context
Anchor	PDF pp9-10 Table1 and section7.3 Figure12
Interpretation limit	Same-system contextual result; source calls this MPS-like non-strict isolation. Not a fault-isolation guarantee or general hardware result.

ONEI-02 P99 TTFT reduction

Field	Extracted value/context
Metric	P99 TTFT reduction
Value	20.7
Unit	percent
Baseline	vLLM
Context	C1 three-model combination; Alpaca/ShareGPT and Azure coding traces; author-reported average across varied arrival rates under non-strict spatial sharing
Anchor	PDF pp9-10 Table1 and section7.3 Figure12
Interpretation limit	Same-system contextual result; source calls this MPS-like non-strict isolation. Not a fault-isolation guarantee or general hardware result.

ONEI-03 Throughput increase

Field	Extracted value/context
Metric	Throughput increase
Value	6.6
Unit	percent
Baseline	vLLM
Context	C1 three-model combination; Alpaca/ShareGPT and Azure coding traces; author-reported average across varied arrival rates under non-strict spatial sharing
Anchor	PDF pp9-10 Table1 and section7.3 Figure12
Interpretation limit	Same-system contextual result; source calls this MPS-like non-strict isolation. Not a fault-isolation guarantee or general hardware result.

Field	Extracted value/context

H.7 S027 / P030 GACER

[33]

Field	Source-based record
Record ID	R00647
Title	GACER: Granularity-Aware ConcurrEncy Regulation for Multi-Tenant Deep Learning
Venue/year	IEEE/ACM International Conference on Computer-Aided Design (ICCAD2024); 2024
DOI	10.1145/3676536.3676718
Source	https://par.nsf.gov/servlets/purl/10643398
Version evidence hash	3cd5b49a18ddaeb2de98c0c0cad1f8850ef1796686d9e79a9b654f00fe78d575
Eligibility	Include; all IC1-IC6 met on inspected primary text
Decision date UTC	2026-10-22T22:37:00.954370+00:00

Criterion	Evidence and anchor
IC1	English final9page ICCAD2024 primary conference paper with exact DOI, copyright and venue citation. PDF p1
IC2	Cross-stack runtime/compiler optimization splits microbatches and reorders operators across concurrent independent DNN model graphs using GPU streams and CPU/GPU synchronization. PDF pp4-6; sections4.2-4.4
IC3	Section5.1 specifies model-inference batches and vision/language/recommendation workloads; training motivation is not substituted for inference results. PDF p7 section5.1
IC4	Normalized batch-completion latency, achieved SM occupancy and optimization costs are reported with comparator descriptions and model settings. PDF pp7-8; Figures7-8; Tables2-4
IC5	Section5.4 explicitly evaluates2,4and5concurrent models on one A100; Table2 provides within-device latencies. P6000/1080Ti outcomes are separate. PDF p8 section5.4 and Tables2,4
IC6	ICCAD2024conference date and indexed2025date are both inside the2020-to-search window. PDF p1; frozenR00647metadata

QA	Score	Rationale and source anchor
Q1	2/2	Multi-tenant resource efficiency and execution-latency objectives are explicit. PDF pp1-2
Q2	2/2	Operator splitting/fusion, stage pointers, stream synchronization and joint heuristic search are described. PDF pp4-6; sections4.2-4.4
Q3	1/2	Workload sizes and profiling procedure are described, but trials/seeds, confidence intervals and warm-up details are not established. PDF pp6-8; Algorithm1 and section5
Q4	1/2	GPU models, model types, datasets, batch/image sizes are stated; complete software/driver versions and device-memory configurations are not supplied. PDF p7 section5.1; p8 Table4
Q5	1/2	Sequential, stream-parallel and MPS-style comparators are defined. Static FLOPS-based sharing and CNN-only AutoMT support constrain fairness across heterogeneous model mixes. PDF p7 sections5.1-5.2 and Figure7
Q6	2/2	Achieved SM occupancy is measured by Nsight; latency speedups and tenant-count scaling are inspectable. Occupancy is not GPU busy-time utilization. PDF pp7-8; Figure8 and Table2
Q7	1/2	Batch-completion latencies are extractable; per-tenant service tails/SLO/load-arrival measurements are not established. PDF pp7-8 Figure7 and Table2
Q8	2/2	Concurrent model mixes,2/4/5tenants, batch variation and three GPU types test co-location behavior. PDF pp7-8 sections5.1-5.6
Q9	N/A	No resource/security/fault-isolation guarantee is claimed. Stage synchronization and occupancy optimization do not establish such a boundary; protocol permits reasonedN/A for unclaimed criteria. PDF sections4-6; no isolation claim/test identified
Q10	1/2	Algorithms, profiling and overhead dependence aid informed reproduction; complete artifact/environment and repeated-trial methodology are not provided. PDF pp5-8 Algorithm1 and Tables3-4

GACER-01 Batch-completion latency

Field	Extracted value/context
Metric	Batch-completion latency
Value	3.3
Unit	ms

Field	Extracted value/context
Baseline	CuDNN-Seq4.32ms; TVM-Seq4.20ms; Stream-Parallel3.81ms; Competitive-Sharing3.89ms
Context	OneA100; two concurrent modelsVGG16+LSTM; Table2 reports1.31x speedup versus sequential
Anchor	PDF p8 Table2
Interpretation limit	Batch completion is not per-request p99 or a serviceSLO.

GACER-02 Batch-completion latency

Field	Extracted value/context
Metric	Batch-completion latency
Value	22.14
Unit	ms
Baseline	CuDNN-Seq37.86ms; TVM-Seq32.24ms; Stream-Parallel25.31ms; Competitive-Sharing27.22ms
Context	OneA100; five concurrent modelsVGG+R18+R34+BERT+DLRM; source reports1.71x versus sequential
Anchor	PDF p8 Table2
Interpretation limit	Table2labelsVGGwithoutversion; do not silently normalize it toVGG16.

GACER-03 Optimization cost range

Field	Extracted value/context
Metric	Optimization cost range
Value	0.84-1.26
Unit	s
Baseline	No comparator for this overhead table
Context	Table3three model mixes, batch sizes4to128; heuristic/search optimization overhead
Anchor	PDF p8 Table3 and section5.5

Field	Extracted value/context
Interpretation limit	Seconds of optimization are not request latency; online suitability is conditional on tolerated reconfiguration delay.

GACER-04 Batch-completion latency

Field	Extracted value/context
Metric	Batch-completion latency
Value	8.51
Unit	ms
Baseline	CuDNN-Seq14.50ms; Stream-Parallel10.13ms
Context	OneGTX1080Ti; R34+LSTM+BST; visionbatch8; source reports1.70x versus sequential
Anchor	PDF p8 Table4 and section5.6
Interpretation limit	Different hardware and workload fromA100Table2; do not pool as matched comparison.

I Continuation full-text exclusions

Only assessed reports enter this table. Pending or inaccessible records are not relabeled as scientific exclusions.

Record / DOI	Decisive reason
R02352; `10.32604/cmc.2026.077259`	EC2: The evaluated intervention is cross-venue mobile-agent reasoning, routing and policy-gated summaries, not an evaluated GPU-sharing mechanism. Four RTX3090venues and16concurrent agents are reported, but no GPU partition/scheduler/remoting comparison or extractable per-GPU co-tenant sharing intervention is established. Evidence: PDF pp8-12 sections4-5.3
R03930; `10.32604/cmc.2025.059528`	EC2: The intervention predicts social-media repost behavior using semantic/temporal model adapters. Methods and results evaluate accuracy/F1/MAPE/MRSE. One V100 is used for model training; there is no concurrent inference GPU-sharing evaluation. Evidence: PDF pp9,11-13 sections4-5

J Headings 23–26 completion map

Heading	Requirement	Current evidence	Remaining qualification
23	25mandatory deliverables	The inherited crosswalk plus the current inclusion/QA/extraction appendices and separate electronic registers	Prepared report/evidence; final whole-review PRISMA and eligibility closure remain open
24	13main report sections	Introduction; RQs; Method; Selection; Taxonomy; RQ synthesis; Comparison; Trade-offs; Discussion; Recommendation; Gaps; Validity; Conclusion	All13sections retained and seven additions integrated
25	Analytical standard	Mechanism taxonomy; contextual comparative matrix; adverse outcomes; trade-offs; fiveRQ answers; conditional recommendations	Analysis integrated; systematic selection remains incomplete at full text
26	Research quality	Stable IDs, primary URLs/hashes, DOI references, eligibility reasons, QA/extraction anchors and explicit limits	Traceability prepared; historical search reconstruction and325advanced records remain open